

CHAPTER

5

Description of Maintenance Service Level

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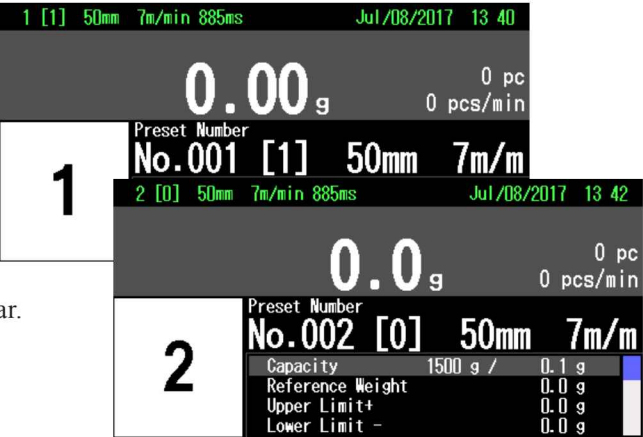
5.1 Description of Level 3

In the Level 3, drive parameters and display languages can be changed by unit self-diagnosis, weigh sensor status check, changing drive unit oalameters by changing conveyorand, and language setting. Presetting values and other set values can be initialized, too. Here describes the parameters displayed on the screen.

5.1.1 Screen Display of Level 3

1. Turn the main power switch ON.

- Zero adjustment screen will appear.
- Main menu screen shown at right will appear.



2. Press .

- Setup Menu screen will appear.



3. Point the cursor to [Access Level Change], and press .

- Screen to input a password will appear.



4. Input the password of the Level 3.



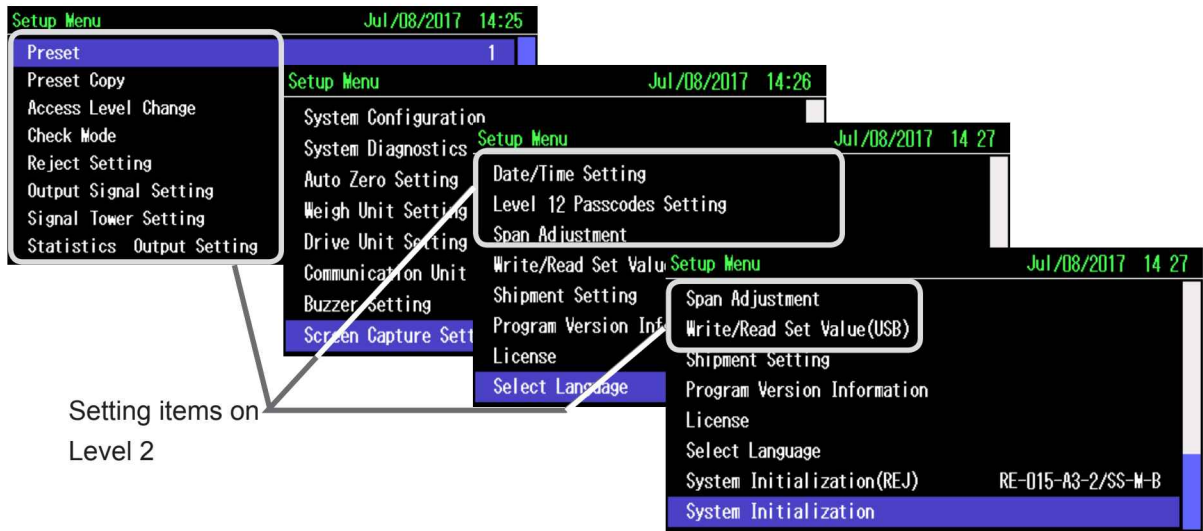
NOTE

- Contact to ISHIDA agent for the password required in the Maintenance Service level.
- When entering the password, the numbers are displayed as "???" on keypad.



5 Description of Maintenance Service Level

• Setup Menu screen at the Level 3 will appear.

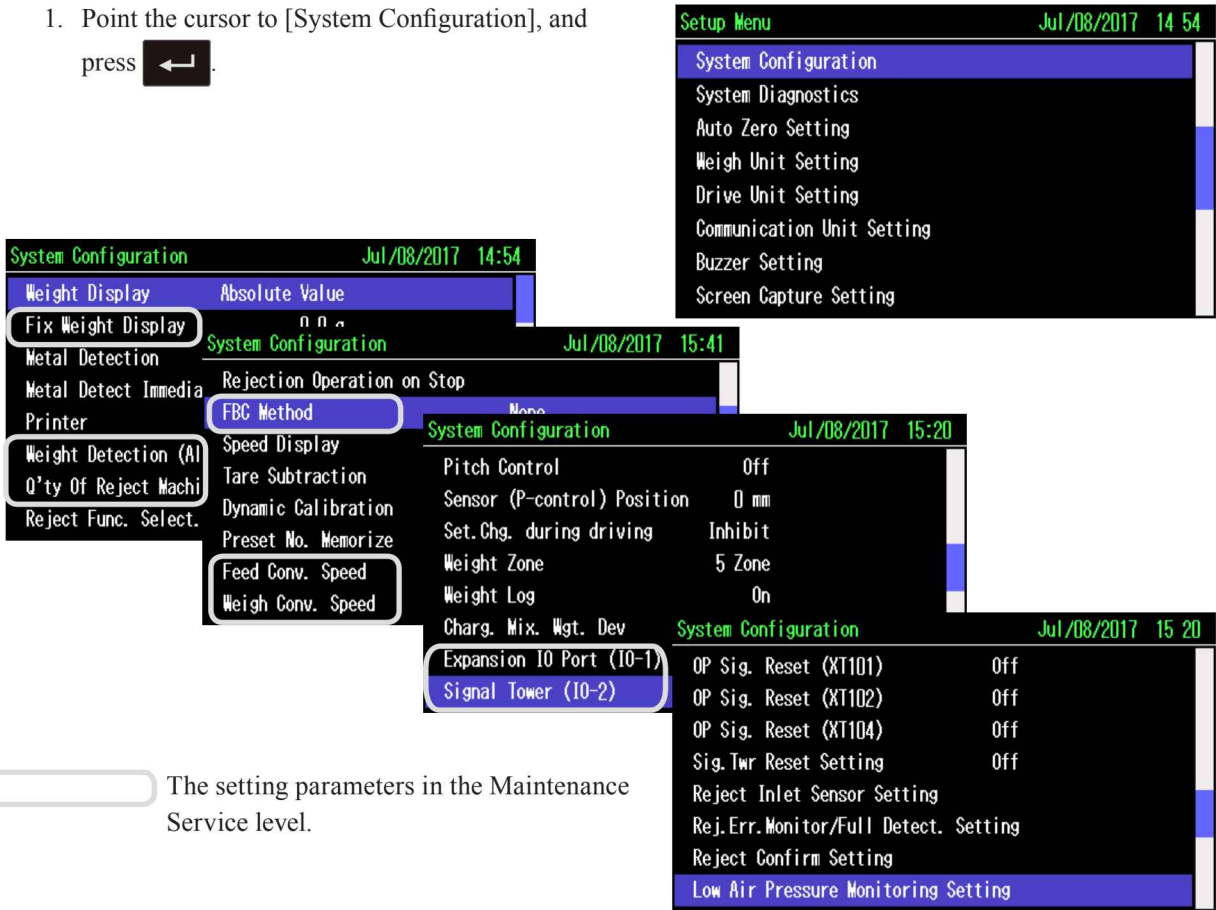


For the parameters settable on the screen in the Level 2, refer to the Instruction Manual.

5.1.2 System Configuration

This section explains which items can be set at the Level 3 in the range of the system configuration setting.

- 1. Point the cursor to [System Configuration], and press



■ Setting items on Level 2

System Configuration		Jul/08/2017 15:20
Weight Display(Standby)	On	
Auto Zero Adjustment	Off	
Test Mode	Off	
Static Filter Number	99	
Input Signal Port Setting		
Input Signal Setting		
Ext. 1 Ip Sig Name Edit	Ext.1	
Ext. 2 Ip Sig Name Edit	Ext.2	

System Configuration		Jul/08/2017 15 21
Input Signal Port Setting		
Input Signal Setting		
Ext. 1 Ip Sig Name Edit	Ext.1	
Ext. 2 Ip Sig Name Edit	Ext.2	
User Log In	Off	
Startup Check	Off	
Standby Disp.Pref.	Enlarge Weight	
Select Language on Startup	On	



Which items can be set at the setup Level 2 in the range of the system configuration setting is listed in the instruction manual. Please refer to it.

5.1.2.1 Fix Weight Display

Displays the fix weight for each model.

1. Point the cursor to [Fix Weight Display], and press .

• [Fix Weight Display] menu will appear.

System Configuration		Jul/08/2017 15 26
Weight Display	Absolute Value	
Fix Weight Display	0.0 g	
Metal Detection	Off	
Metal Detect Immediate Stop	Off	
Printer	Off	
Weight Detection (All Preset)	On	
Q'ty Of Reject Machine	1 pc	
Reject Func. Select.	For All	

2. Set the values with the ten-key pad, press .

System Configuration		
Fix Weight Display	0.0 g	Jul/08/2017 15:26
	0.0	



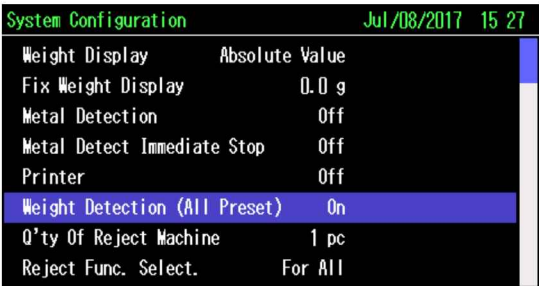
The default value is set to "0.0" (g). (It depends on the model or specification.)

5 Description of Maintenance Service Level

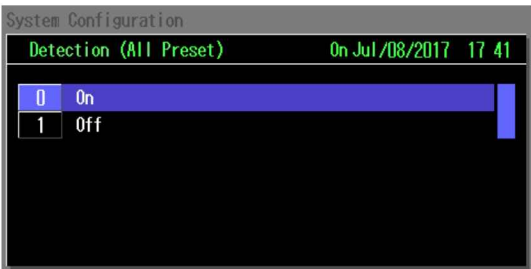
5.1.2.2 Weight Detection (All Preset)

The Weight Detection is used to temporarily skip the weight inspection, when the load cell is damaged for instance. Setting to "Off" skips the judgment of correct weight, overweight, and underweight, and the products are conveyed without the weight inspected.

1. Point the cursor to [Weight Detection (All Preset)], and press  .
 - [Weight Detection (All Preset)] menu will appear.
2. Press   to select one out of "on" or "off", and press  .



System Configuration		Jul/08/2017 15 27
Weight Display	Absolute Value	
Fix Weight Display	0.0 g	
Metal Detection	Off	
Metal Detect Immediate Stop	Off	
Printer	Off	
Weight Detection (All Preset)	On	
Q'ty Of Reject Machine	1 pc	
Reject Func. Select.	For All	



System Configuration		On Jul/08/2017 17 41
Detection (All Preset)		
0	On	
1	Off	

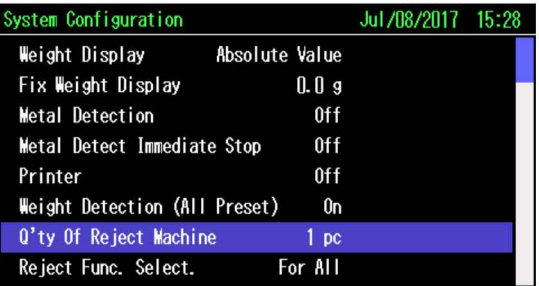


- **Normally set to "On". Setting to "Off" skips the judgment of correct weight, overweight, and underweight, and the products are conveyed without the weight inspected.**
- **The inspection signals of metal detection and X-ray inspection are not influenced by the setting of Weight Detection.**

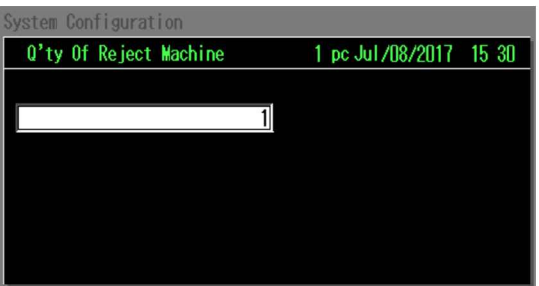
5.1.2.3 Q'ty of Reject Machine

Specify the number of reject machine (RE) to use.

1. Point the cursor to [Q'ty of Reject Machine], and press  .
 - [Q'ty of Reject Machine] menu will appear.
2. Set the values with the ten-key pad, press  .



System Configuration		Jul/08/2017 15:28
Weight Display	Absolute Value	
Fix Weight Display	0.0 g	
Metal Detection	Off	
Metal Detect Immediate Stop	Off	
Printer	Off	
Weight Detection (All Preset)	On	
Q'ty Of Reject Machine	1 pc	
Reject Func. Select.	For All	



System Configuration		1 pc Jul/08/2017 15 30
Q'ty Of Reject Machine		
1		



The standard model has been set to "1".

5.1.2.4 FBC Method (Feed Back Control) (optional)

This option is displayed when the Feedback Control (FBC) is activated by License Attestation.
Sets the Feed back control system.

1. Point the cursor to [FBC Method], and press



• [FBC Method] menu will appear.

System Configuration		Jul/08/2017 15:41
Rejection Operation on Stop		
FBC Method	None	
Speed Display	Detailed	
Tare Subtraction	Off	
Dynamic Calibration	Off	
Preset No. Memorize	On	
Feed Conv. Speed	Variable1	
Weigh Conv. Speed	Variable1	

2. Press key to select FBC Method, and press .

- 1: Indirect ctrl pulse
- 2: Indirect ctrl wei
- 3: Direct ctrl
- 4: Indirect ctrl tend

Select from the above methods.

System Configuration		
FBC Method		None Jul/08/2017 15:42
0	None	
1	Indirect Ctrl Puls	
2	Indirect Ctrl Wei	
3	Direct Ctrl	
4	Indirect Ctrl Tend	

3. Point the cursor to "1 Indirect ctrl puls", and press .

System Configuration		
FBC Method		None Jul/08/2017 15:45
0	None	
1	Indirect Ctrl Puls	
2	Indirect Ctrl Wei	
3	Direct Ctrl	
4	Indirect Ctrl Tend	

- The screen shows that the option [Indirect ctrl pulse] is selected.

System Configuration		Jul/08/2017 15:53
Rejection Operation on Stop		
FBC Method	Indirect Ctrl Puls	
FBC	Single	
FB Output Mode	Off	
Speed Display	Detailed	
Tare Subtraction	Off	
Dynamic Calibration	Off	
Preset No. Memorize	On	



Since the standard model does not use FBC, "0 None" is set.



For Communication Unit Setting, refer to "7.5.3 RS-232C setting information".

5.1.2.5 Feed Conveyor Speed

Sets the infeed conveyor speed control.

1. Point the cursor to [Feed Conv. Speed], and press



- [Feed Conv. Speed] menu will appear.

2. Press key to select one out of

"Fixed", or "Variable 1", and press .

System Configuration		Jul/08/2017 15:37
Rejection Operation on Stop		
FBC Method	None	
Speed Display	Detailed	
Tare Subtraction	Off	
Dynamic Calibration	Off	
Preset No. Memorize	On	
Feed Conv. Speed	Variable1	
Weigh Conv. Speed	Variable1	

System Configuration		
Feed Conv. Speed		Variable1 Jul/08/2017 17:04
0	Fixed	
1	Variable1	



- Since the standard model mounts the infeed conveyor, "Variable 1" is initially set.
- In removing the infeed conveyor, set to "Fixed".

5.1.2.6 Weigh Conveyor Speed

Sets the weigh conveyor speed control.

1. Point the cursor to [Weigh Conv. Speed], and press .

- [Weigh Conv. Speed] menu will appear.

2. "Variable 1" will appear.

System Configuration		Jul/08/2017 17:05
Rejection Operation on Stop		
FBC Method	None	
Speed Display	Detailed	
Tare Subtraction	Off	
Dynamic Calibration	Off	
Preset No. Memorize	On	
Feed Conv. Speed	Variable1	
Weigh Conv. Speed	Variable1	

Weigh Conv. Speed		Variable1 Jul/08/2017 17:05
1	Variable1	



Standard setting is "Variable 1".

5.1.2.7 Rejection Conveyor Speed

Sets the rejection conveyor speed control.

1. Point the cursor to [Rejection Conv. Speed], and press .

• [Rejection Conv. Speed] menu will appear.

System Configuration		Jul/08/2017 17 06
Reject Conv. Speed	Variable1	
Pitch Control	Off	
Sensor (P-control) Position	0 mm	
Reference Value Update Setting		
Set.Chg. during driving	Inhibit	
Weight Zone	5 Zone	
Weight Log	On	
Charg. Mix. Wgt. Dev	Off	

2. Press   to select one out of "Fixed" or "Variable 1", and press .

System Configuration		Jul/08/2017 17 06
Reject Conv. Speed	Variable1	
0	Fixed	
1	Variable1	



Since the standard model does not mount the rejector, "Fixed" is initially set. When the rejecting conveyor is mounted, "Variable 1" is set.

5.1.2.8 Expansion IO port (IO-1) (optional)

The expansion IO port enables signal communication with other interlocking machines (used when the machine is interlocked with metal detection machines).

1. When the expansion IO port is attached, "On" is automatically displayed.

System Configuration		Jul/08/2017 17 07
Expansion IO Port (IO-1)	On	
Signal Tower (IO-2)	4stg	
OP Sig. Reset (XT101)	Off	
OP Sig. Reset (XT102)	Off	
OP Sig. Reset (XT104)	Off	
Sig.Twr Reset Setting	Off	
Reject Inlet Sensor Setting		
Rej.Err.Monitor/Full Detect. Setting		

2. When the expansion IO port is not attached, "Off" is automatically displayed.

System Configuration		Jul/08/2017 17 07
Expansion IO Port (IO-1)	On	
Signal Tower (IO-2)	4stg	
OP Sig. Reset (XT101)	Off	
OP Sig. Reset (XT102)	Off	
OP Sig. Reset (XT104)	Off	
Sig.Twr Reset Setting	Off	
Reject Inlet Sensor Setting		
Rej.Err.Monitor/Full Detect. Setting		



The expansion IO port is not attached to the standard model, and the setting is fixed to "Off" and cannot be changed (grayed out).
When the model is optionally attached with the expansion IO port, the setting is fixed to "On" and cannot be changed (grayed out).

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5.1.2.9 Signal Tower (IO-2) (optional)

Set either "2stg", "4stg" or "6stg" for the signal tower.

1. If no signal tower is equipped, "Off" is displayed.

- If a signal tower is equipped, "4stg" is displayed.

System Configuration		Jul/08/2017 17:23
Expansion IO Port (IO-1)	Off	
Signal Tower (IO-2)	Off	
OP Sig. Reset (XT101)	Off	
Reject Inlet Sensor Setting		
Rej.Err.Monitor/Full Detect. Setting		
Reject Confirm Setting		
Low Air Pressure Monitoring Setting		
Weight Display(Standby)	On	

2. Point the cursor to [Signal Tower (IO-2)], and

press .

System Configuration		Jul/08/2017 17:19
Expansion IO Port (IO-1)	Off	
Signal Tower (IO-2)	4stg	
OP Sig. Reset (XT101)	Off	
Sig.Twr Reset Setting	Off	
Reject Inlet Sensor Setting		
Rej.Err.Monitor/Full Detect. Setting		
Reject Confirm Setting		
Low Air Pressure Monitoring Setting		

- [Signal Tower (IO-2)] menu will appear.

3. Press   to select "2stg", "4stg" or

"6stg" and press .

Signal Tower (IO-2)		4stg	Jul/08/2017 17:20
1	4stg		
2	6stg		
3	2stg		



Since the standard model does not mount the signal tower, "OFF" is initially set. When the optional signal tower is used, set to "2stg", "4stg" or "6stg".

5.1.2.10 Rejection Error Monitoring/Full Detect Setting (optional)

Rejection Error Monitoring/Full Detect Setting function is the function to monitor the defective weight products are rejected correctly by mounting photo eye unit on the defective weight direction of the rejection conveyor.

1. Point the cursor to [Rej. Err. Monitor/Full Detect. Setting], and press .

- [Rej. Err. Monitor/Full Detect. Setting] screen will appear.

System Configuration		Jul/08/2017 17:34
Expansion IO Port (IO-1)	On	
Signal Tower (IO-2)	4stg	
OP Sig. Reset (XT101)	Off	
OP Sig. Reset (XT102)	Off	
OP Sig. Reset (XT104)	Off	
Sig.Twr Reset Setting	Off	
Reject Inlet Sensor Setting		
Rej.Err.Monitor/Full Detect. Setting		

2. Press .

- [Rej. Dir. sensor Use] screen will appear.

Rej.Err.Monitor/Full Detect. Setting		Jul/08/2017 17:35
Rej.Dir.Sensor Use	No Sensor	

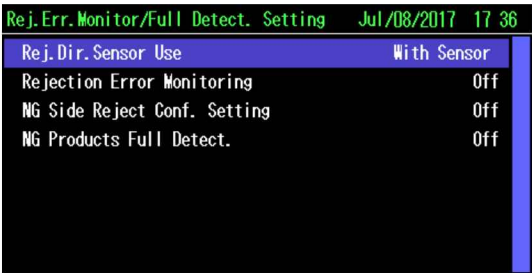
3. Press   to select one out of [Dir. 1 Sensor Use] or [Dir. 2 Sensor Use], and press .



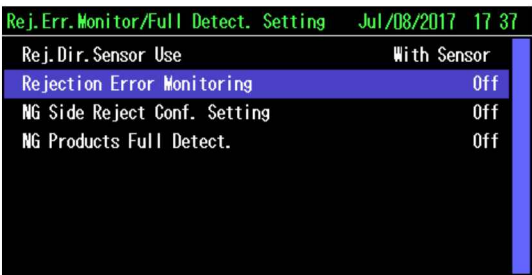
4. Press  to select "On" or "Off".

5. Press .

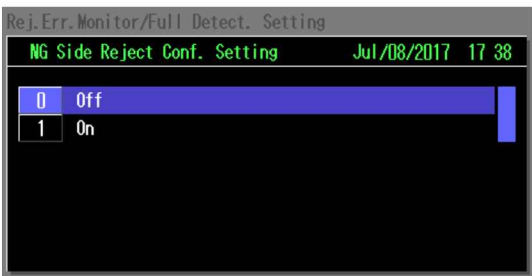
- Return to [Rej. Dir. Sensor Use] screen



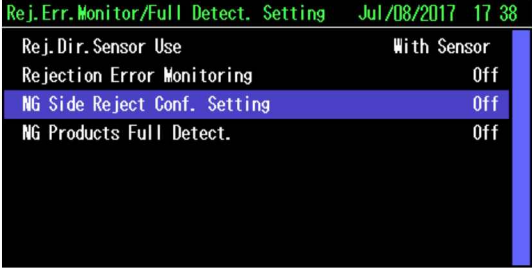
6. Select [Rejection Error Monitoring], and press .



7. Press   to select "On" or "Off", and press .

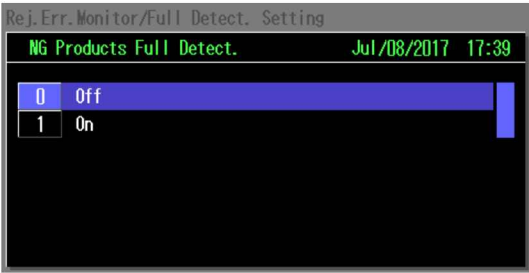


8. Point the cursor to [NG Side Reject Conf. Setting], and press .



5 Description of Maintenance Service Level

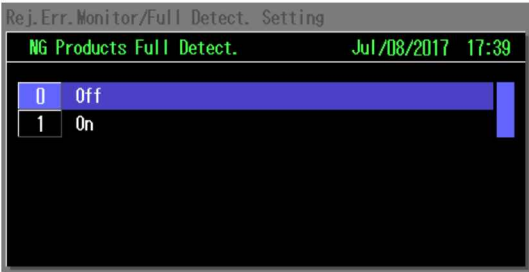
9. Press   to select "On" or "Off", and press .



10. Point the cursor to [NG Products Full Detect.], and press .



11. Press   key to select "On" or "Off", and press .



12. Press .
- Return to [Rej. Err. Monitor/Full Detect. Setting] screen.



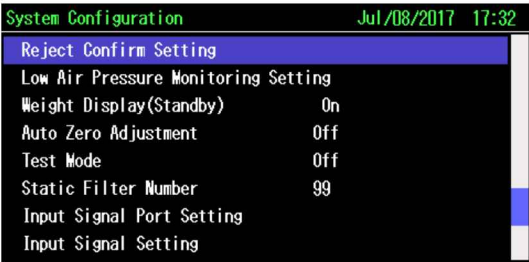
The standard model does not support the Err. Monitor/Full Detect function, and it is set to "Off". When the model is optionally attached with the Err. Monitor/Full Detect function, select [Dir.1 Sensor Use] and [Dir.2 Sensor Use], and attach the photoelectric sensor to the directions 1 and 2 of the rejector.

5.1.2.11 Reject Confirm Setting (optional)

The Reject Confirm Setting function detects that no defective product passes and the products of correct weight pass without problem, with the photoelectric sensor attached at the rejector in the product (correct weight) discharge side.

1. Point the cursor to [Reject Confirm Setting], and press the .

- [Reject Confirm] menu will appear.



2. Press   to select one out of "On" or "Off", and press .



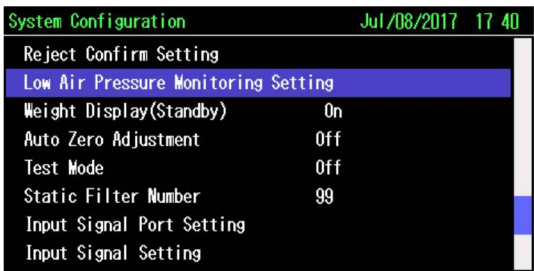
As the standard model has no transfer error detection setting, it is set to "Off". When the model is optionally equipped with the transfer error detection setting, set to "On", and attach the photoelectric sensor to the product (correct weight) discharge side of the rejector.

5.1.2.12 Low air pressure monitoring Setting (optional)

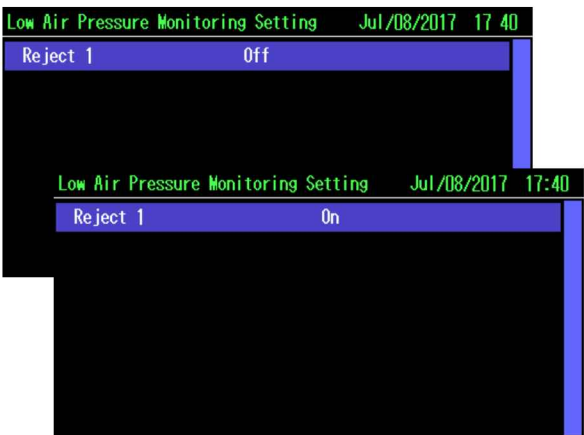
"Low Air Pressure Monitoring" is a function to output the monitoring signal when the air pressure falls below the pressure sensor setting value.

1. Point the cursor to [Low Air Pressure Monitoring Setting], and press the .

- [Low Air Pressure Monitoring] menu will appear.



2. Press  to select one out of "On" or "Off",



This Low Air Pressure Monitoring function is not equipped with the standard model, and this function is set to OFF. When the Low Air Pressure Monitoring is provided as an option, set the setting to ON and attach the pressure sensor to the air regulator.

5.1.2.13 Weight display (standby)

Sets the weight display during standby, "ON" or "OFF".

1. Point the cursor to [Weight display (standby)], and press .

• [Weight display (standby)] menu will appear.

System Configuration		Jul/08/2017 17:41
Reject Confirm Setting		
Low Air Pressure Monitoring Setting		
Weight Display(Standby)	On	
Auto Zero Adjustment	Off	
Test Mode	Off	
Static Filter Number	99	
Input Signal Port Setting		
Input Signal Setting		

2. Press   key to select one out of "On" or "Off", and press .

System Configuration		On Jul/08/2017 17 41
Weight Display(Standby)		
0	On	
1	Off	



"ON" is initially set. Set to "OFF" when this function is not used.

5.1.2.14 Auto Zero Adjustment

Auto Zero Adjustment is the function that performs zero adjustment automatically when the power tuens ON.

1. Point the cursor to [Auto Zero Adjustment], and press .

• [Auto Zero Adjustment] menu will appear.

System Configuration		Jul/08/2017 17:42
Reject Confirm Setting		
Low Air Pressure Monitoring Setting		
Weight Display(Standby)	On	
Auto Zero Adjustment	Off	
Test Mode	Off	
Static Filter Number	99	
Input Signal Port Setting		
Input Signal Setting		

2. Press   to select one out of "On" or "Off", and press .

System Configuration		Off Jul/08/2017 17 42
Auto Zero Adjustment		
0	Off	
1	On	

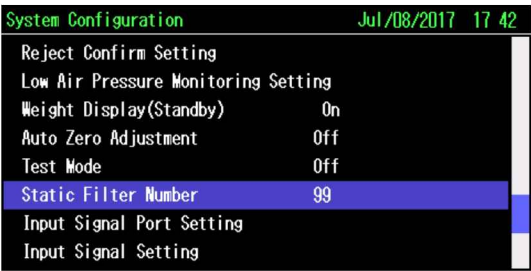


Since the standard model is not use this function, "OFF" is initially set. Set to "ON" when perform zero adjustment when the power turns ON.

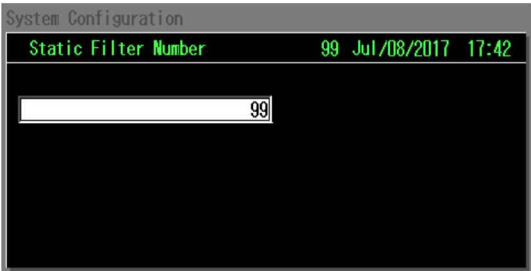
5.1.2.15 Static Filter Number

Displays the static filter number.

1. Point the cursor to [Static Filter Number], and press  .
 - [Static Filter Number] menu will appear.



2. Input two digits, and press  .

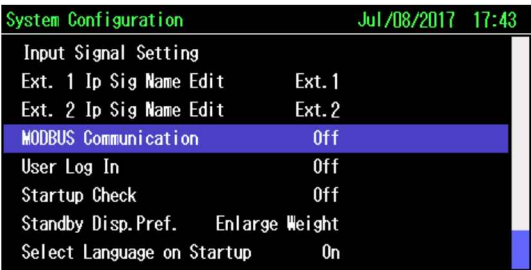


The default is "99". "99" means that the same filter is used as operation.

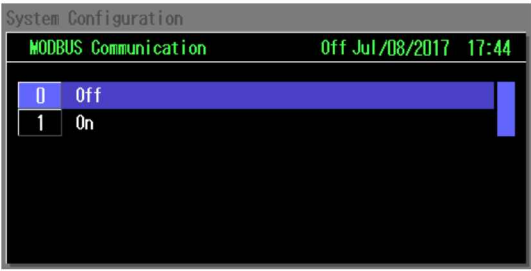
5.1.2.16 MODBUS Communication (optional)

This option is displayed when the host interlocking operation has been attested by License Attestation. It switches MELSEC Communication and MODBUS Communication. When [MODBUS Communication] is set to "Off", the system selects the default [MELSEC Communication].

1. Point the cursor to [MODBUS Communication], and press  .
 - [MODBUS Communication] menu will appear.



2. Press   key to select of "On" or "Off", and press  .



Under the default setting, MELSEC Communication has been selected where MODBUS Communication is "Off". Set to "On" as necessary depending on the specification.

5.1.2.17 After changing the details in system configuration

After changing the detailed parameters in the system configuration, a message "Completed. Turn off the power and restart" appears on the screen. The setting data is updated when the power is turned ON again. Follow the prompts on the screen.

1. After changing the details of the system configuration, press the  .
 - A message "Completed. Turn off the power and restart" appears on the screen.
2. Turn ON the power again according to the prompts on the screen.



5.1.3 System Diagnostics

System Diagnostics enables such functions as Operation Unit Diagnosis (diagnosis of Numeric keypad, Display Unit, and USB memory), Auxiliary Unit Diagnosis (Detailed Weigh Data Display, Rejector Automatic/Manual Test, Input Port Status Display), and conveyor separate operation.

1. Point the cursor to [System Diagnostics], and press  .
 - Selection screens for diagnosis of operation unit and of auxiliary unit, and for conveyor separate operation.



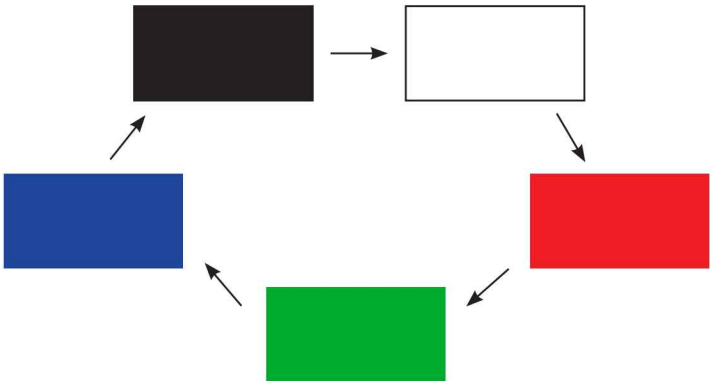
5.1.3.1 Operation Unit Diagnosis

1. Point the cursor to [Operation Unit Diagnosis], and press  .
 - [Operation Unit Diagnosis] screen will appear.



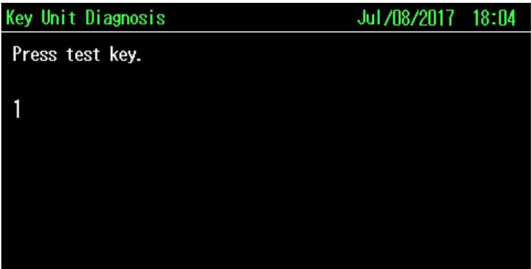
■ Operation Unit Diagnosis

- Judges whether the displayed screen is correct or not.
1. Point the cursor to [Display Unit Diagnosis], and press  .
- Displays the lighted/lights-out displays by turns.



■ Key unit diagnosis

- Diagnoses whether operation keys function properly.
1. Point the cursor to [Key Unit Diagnosis], and press  .
2. Press any key.
 - The input character is displayed on the screen.



NOTE If there is no response when pressing the particular key, examine whether the key board is damaged.

■ Printer unit diagnosis (option)

- Perform the printer unit diagnosis in the following steps.
1. Point the cursor to [Printer unit diagnosis], and press  .
- The message "Print testing." appears on the screen. Pressing any key on the screen starts the printer diagnosis.



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5 Description of Maintenance Service Level

- Pressing the  key can cancel the diagnosis.



To activate the printer unit diagnosis, set the following parameters:

- a. [Setup Menu] → [Statistics Output Setting] → [Output Destination] → From [USB] to [Printer].

Setup Menu Jul/08/2017 22:01	
Preset	1
Preset Copy	
Access Level Change	Level 3
Check Mode	
Reject Setting	
Output Signal Setting	
Signal Tower Setting	
Statistics Output Setting	

Statistics Output Setting Jul/26/2017 16:20	
Output Destination	USB
Line Name	0
Total Automatic Change	On
Total Items	All
Total Auto Output	Off
Batch Analysis	Off
Batch Total Auto Output	Off
Batch Total Time	0 h

Statistics Output Setting Jul/26/2017 16:20	
Output Destination	USB
0	Printer
1	USB

Statistics Output Setting Jul/26/2017 16:21	
Output Destination	USB
0	Printer
1	USB

Statistics Output Setting Jul/26/2017 16:21	
Output Destination	Printer
Line Name	0
Total Automatic Change	On
Total Items	All
Total Auto Output	Off
Batch Analysis	Off
Batch Total Auto Output	Off
Batch Total Time	0 h

Operation Unit Diagnosis Jul/10/2017 11:15	
Printer Unit Diagnosis	
Print testing.	
Press any key.	
Press  [Exit] key to cancel.	

- b. [Setup Menu] → [System Configuration] → [Printer] → From [Off] to [On].

Setup Menu Jul/26/2017 16:22	
System Configuration	
System Diagnostics	
Auto Zero Setting	
Weigh Unit Setting	
Drive Unit Setting	
Communication Unit Setting	
Buzzer Setting	
Screen Capture Setting	

System Configuration Jul/26/2017 16:22	
Weight Display	Absolute Value
Fix Weight Display	0.0 g
Metal Detection	Off
Metal Detect Immediate Stop	Off
Printer	Off
Weight Detection (All Preset)	On
Q'ty Of Reject Machine	1 pc
Reject Func. Select.	For All

System Configuration Jul/26/2017 16:23	
Printer	Off
0	Off
1	On

System Configuration Jul/26/2017 16:23	
Weight Display	Absolute Value
Fix Weight Display	0.0 g
Metal Detection	Off
Metal Detect Immediate Stop	Off
Printer	On
Weight Detection (All Preset)	On
Q'ty Of Reject Machine	1 pc
Reject Func. Select.	For All

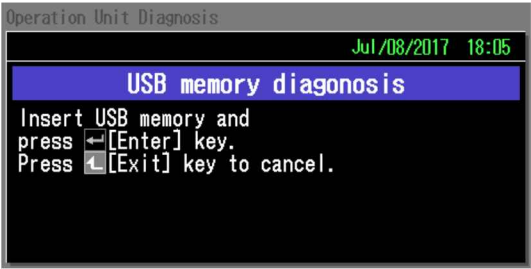
■ USB Memory unit diagnosis

Diagnosis whether the memory data stored in the USB memory function properly.

1. Point the cursor to [USB Memory Diagnosis],
and press  .
 - Screen to insert the USB memory will appear.



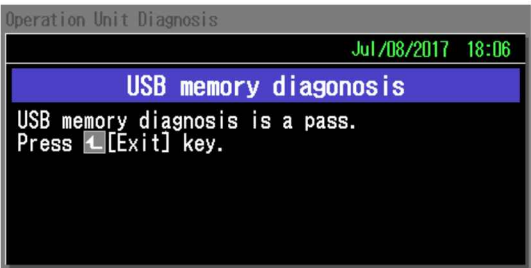
2. Press  to start the USB Memory Diagnosis.
When the diagnosis is not required, press  to return to [USB memory diagnosis] screen.



- On the [USB memory diagnosis] screen, a message "Checking USB memory. Please wait." will appear.

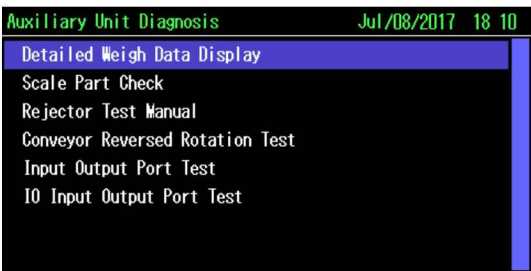


3. When this message is displayed, press  to return to [USB memory diagnosis] screen.



5.1.3.2 Auxiliary Unit Diagnosis

1. Point the cursor to [Auxiliary Unit Diagnosis] on the [System Diagnostics] screen, and press  .
 - [Auxiliary Unit Diagnosis] menu will appear.

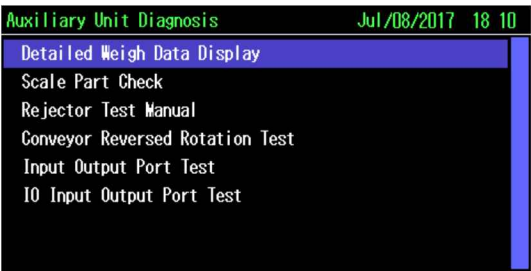


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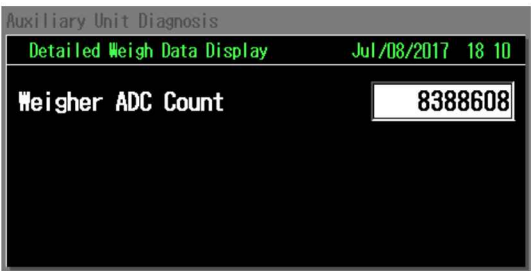
■ Detailed weigh data display

The Detailed Weigh Data Display displays the AD count value, which digitally converts the force balance sensor output on the ADC board and then filtered.

1. Point the cursor to [Detailed Weigh Data Display] screen, and press .



- Detailed weigh data of the selected channel will appear.



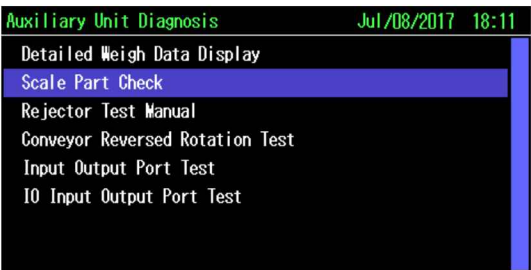
The screen displays the AD count value digitally converting the force balance sensor output on the ADC board and then filtered. Under the normal condition, the AD count is 8388608 at minimum and 13637778 at maximum.

■ Scale part check

Communication with the scale part (ADC board) can be diagnosed.

1. Point the cursor to [Scale Part Check] screen, and press .

- [ADC Com. Sum Value Err.], [ADC Com. Ret. Processing] are displayed.



2. Check [ADC Com. Sum Value Err.] and [ADC Com. Ret. Processing].



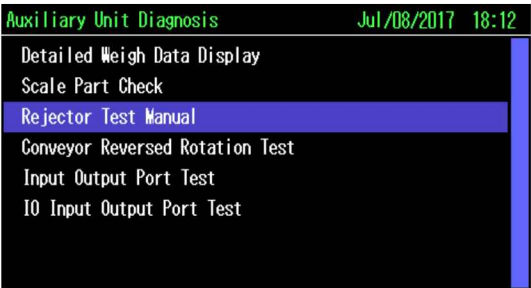
- Press , return to [Auxiliary Unit Diagnosis] screen.

■ **Rejector manual test**

Manually enables the operation check of rejecting mechanism.

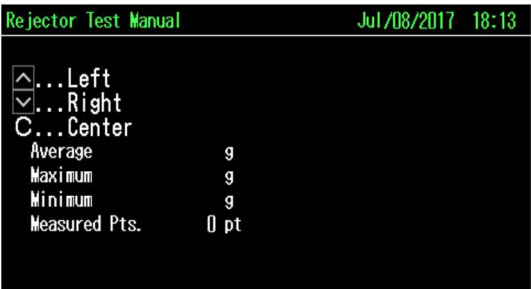
1. Point the cursor to [Rejector Test Manual] and press .

- [Rejector Test: Manual] menu will appear.

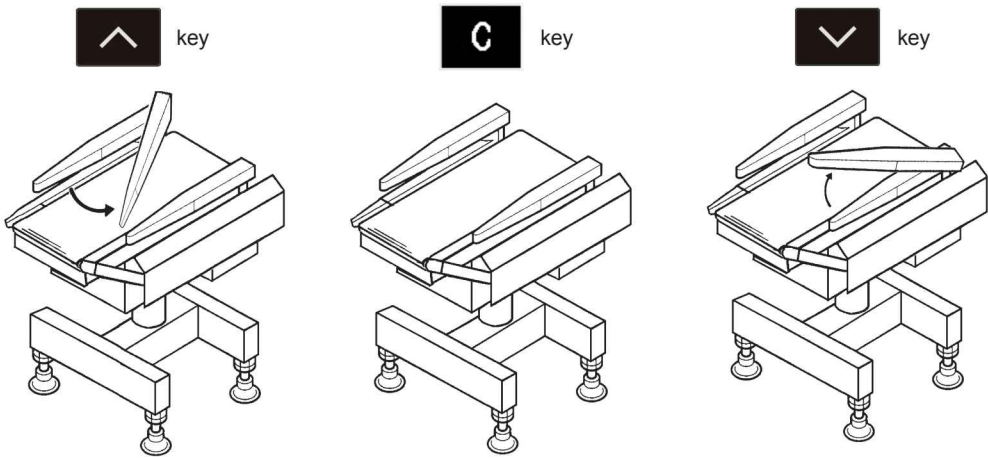


2. Press  to operate the right rejecting arm, and press  to operate the left rejecting arm. Press  key to set them center.

- The rejector of the selected (pressed) direction moves in a cycle.



For the drop-belt type rejector, use only the right direction ( key).

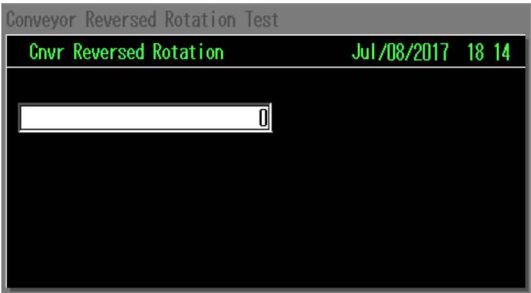
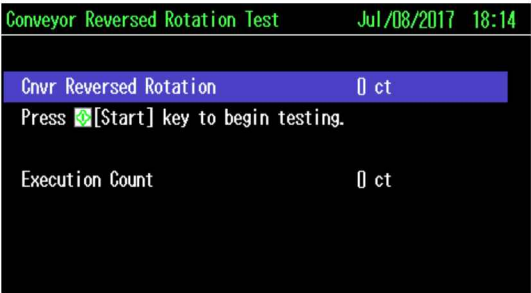
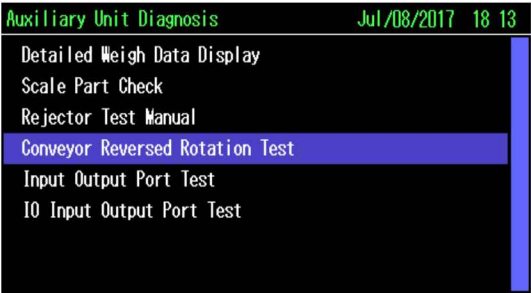


If it is a drop-belt-type rejector, only the "Right direction" is used. The standard model coordinates with the arm rejecting device. Note that the air jet type and dropbelt type may have slightly different operation than described.

■ Conveyor Reversed Rotation test

The conveyor reversed rotation operation can be checked.

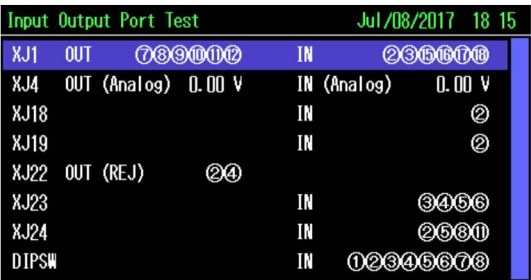
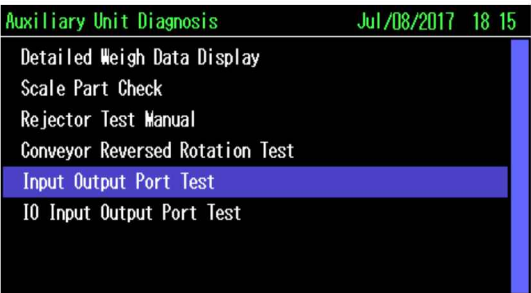
1. Point the cursor to [Conveyor Reversed Rotation Test], and press  .
 - [Conveyor Reversed Rotation Test] menu will appear.
2. Point the cursor to [Cnvr Reversed Rotation], and press  .
 - The normal/reverse rotation count of the infeed conveyor and the weigh conveyor can be arbitrarily set.



■ Input output port test

Enable to check the signal status of the input or output ports.

1. Point the cursor to [Input Output Port Test], and press  .
 - [Input Output Port Test] menu will appear.

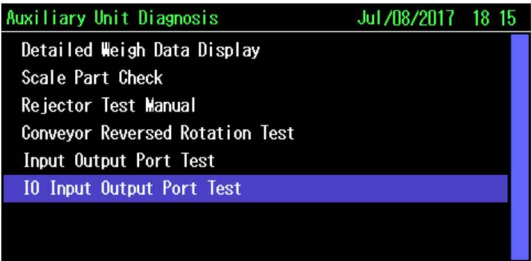


NOTE This function is for only DRV board.

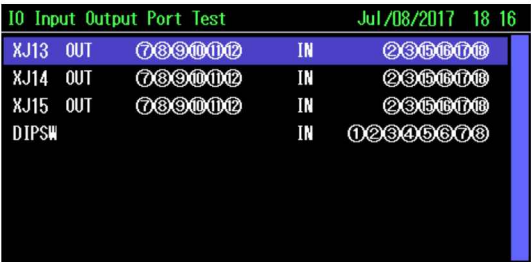
■ IO Input output port test

Enable to check the signal status of the IO input or output ports.

1. Point the cursor to [IO Input Output Port Test],
and press .



- [IO Input Output Port Test] menu will appear.



5.1.3.3 Conveyor Separate Operation

1. Point the cursor to [Conveyor Separate Operation],
and press .

- The weigh conveyor automatically starts the operation.

2. Press  stops the weigh conveyor.



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5.1.4 Automatic Zero Setting

Enables various settings as [Auto Zero], [Auto Zero Weight], [Auto Zero Stable Wt.], [Auto Zero Adjg. Average No.], and [Criteria of Wt. Fluc.]. The initial values depend on weigh capacity.

1. Point the cursor to [Auto Zero Setting], and press



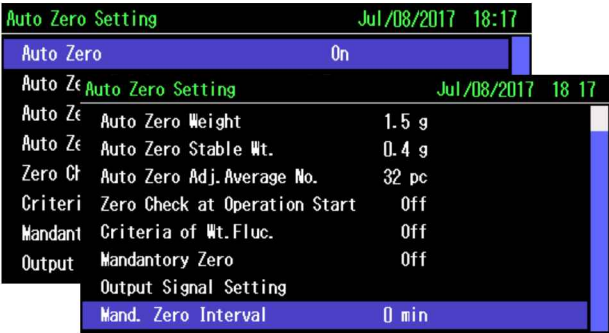
■ Automatic zero setting

Enables to select "On" or "Off" for Auto Zero.

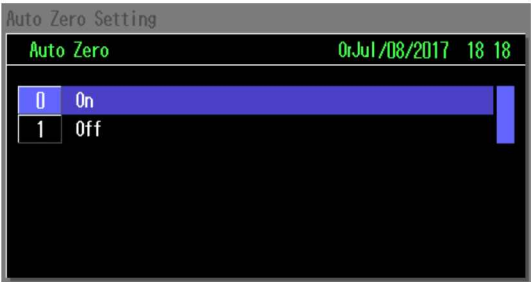
1. Point the cursor to [Auto Zero], and press



- [Auto Zero] menu will appear.



2. Press key to select of "On" or "Off", and press .



■ Auto zero weight

Auto Zero Weight: Enables to set the range in which automatic zero works.

1. Point the cursor to [Auto Zero Weight], and press



- [Auto Zero Weight] menu will appear.



2. Set the values with the ten-key pad, press .

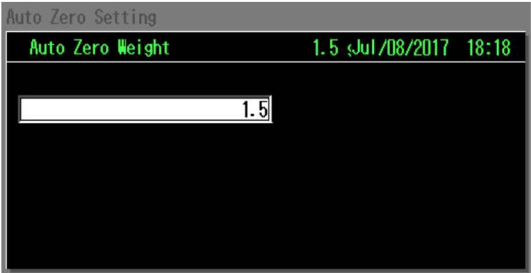


Table for initial values

Weight capacity	S015	S060
Auto ZeroWeight [g]	1.5	6.0
Auto Zero Stable Wt. [g]	0.4	2.0

■ Auto zero stable Wt.

Auto Zero stable Wt.: Enables to set the weight with which the zero point is judged to be stable.

1. Point the cursor to [Auto Zero Stable Wt.], press .

- [Auto Zero Stable Wt.] menu will appear.

2. Set the values with the ten-key pad, press .

Auto Zero Setting		Jul/08/2017 18 19
Auto Zero	On	
Auto Zero Weight	1.5 g	
Auto Zero Stable Wt.	0.4 g	
Auto Zero Adj. Average No.	32 pc	
Zero Check at Operation Start	Off	
Criteria of Wt.Fluc.	Off	
Mandantory Zero	Off	
Output Signal Setting		

Auto Zero Setting		0.4 Jul/08/2017 18 19
Auto Zero Stable Wt.	0.4	

■ Auto zero adj. average No.

Auto Zero Adj. Average No.: Enables to set the average of the set count of zero point data (as the next Auto Zero Adj. Average No.).

1. Point the cursor to [Auto Zero Adj. Average No.], press .

- [Auto Zero Adj. Average No.] menu will appear.

2. Press   key to select of "0-4", and press .

Auto Zero Setting		Jul/08/2017 18:19
Auto Zero	On	
Auto Zero Weight	1.5 g	
Auto Zero Stable Wt.	0.4 g	
Auto Zero Adj. Average No.	32 pc	
Zero Check at Operation Start	Off	
Criteria of Wt.Fluc.	Off	
Mandantory Zero	Off	
Output Signal Setting		

Auto Zero Setting		32 pc Jul/08/2017 18:19
Auto Zero Adj. Average No.	32 pc	
0	16 pc	
1	32 pc	
2	64 pc	
3	128 pc	
4	256 pc	



The default value is set to "1: 32pc". Change to "0" to "4" as necessary.

■ Zero Check at Operation Start

This function checks if the zero point is within the tolerable range at the operation start. If the deviation is over the tolerable range, an error is triggered to disable further operation.

1. Point the cursor to [Zero Check at Operation Start], press .

- [Zero Check at Operation Start] menu will appear.

2. Press   key to select of "on" or "OFF", and press .

Auto Zero Setting		Jul/08/2017 18:20
Auto Zero	On	
Auto Zero Weight	1.5 g	
Auto Zero Stable Wt.	0.4 g	
Auto Zero Adj. Average No.	32 pc	
Zero Check at Operation Start	Off	
Criteria of Wt.Fluc.	Off	
Mandantory Zero	Off	
Output Signal Setting		

Auto Zero Setting		Off Jul/08/2017 18:21
Zero Check at Operation Start	Off	
0	Off	
1	On	



The default value is set to "Off". Change to "On" as necessary.

■ Criteria of Wt. Fluc.

Criteria of Wt. Fluc: Enables to set either "OFF", "0.1% of Weighing", or "1% of Weighing".

1. Point the cursor to [Criteria of Wt. Fluc], and press .
2. Press   key to select of "Off" or 0.1% of Weighing, and press .

Auto Zero Setting		Jul/08/2017 18 21
Auto Zero	On	
Auto Zero Weight	1.5 g	
Auto Zero Stable Wt.	0.4 g	
Auto Zero Adj. Average No.	32 pc	
Zero Check at Operation Start	Off	
Criteria of Wt.Fluc.	Off	
Mandantory Zero	Off	
Output Signal Setting		

Auto Zero Setting		OffJul/08/2017 18 21
Criteria of Wt.Fluc.		
0	Off	
1	0.1% of Weighing	
2	1% of Weighing	



- Under the default setting, "Criteria of Wt. Fluc" has been selected to "OFF". To set a certain value to the criteria, either select "0.1% of Weighing" or "1% of Weighing".
- The Criteria of Wt. Fluc is a function to trigger an error for operation stop when the weighing sensor exceeds 0.1% of the weighing 1% of the weighing, without blocking the photoelectric sensor.

■ Mandantory Zero

This function sets the operation when the weighed products are conveyed after the interval time set for [Mandantory Zero] has elapsed since no auto zero is checked.

1. Point the cursor to [Mandantory Zero], and press .
2. Press   to select of "off", "Signal Out" or "Feed Stop" and press .

Auto Zero Setting		Jul/08/2017 18 21
Auto Zero	On	
Auto Zero Weight	1.5 g	
Auto Zero Stable Wt.	0.4 g	
Auto Zero Adj. Average No.	32 pc	
Zero Check at Operation Start	Off	
Criteria of Wt.Fluc.	Off	
Mandantory Zero	Off	
Output Signal Setting		

Auto Zero Setting		OffJul/08/2017 18 21
Mandantory Zero		
0	Off	
1	Signal Out	
2	Feed Stop	

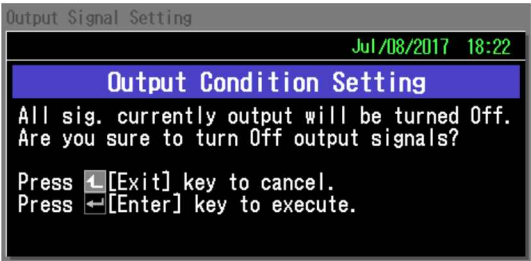
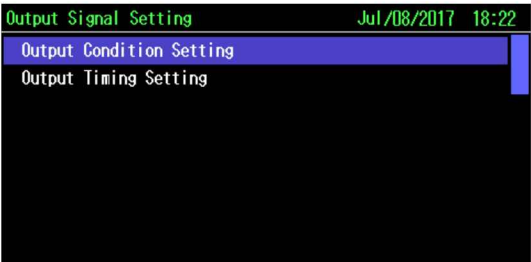
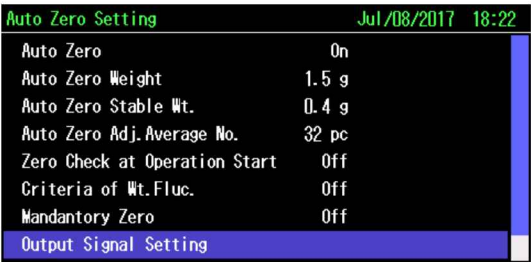


- Off : No operation is given.
Signal Out : Triggers the output signal of Mandantory Zero point.
Feed Stop : Triggers the output signal of Mandantory Zero point, and stops the infeed conveyor after weighing is completed for the time set for [Mand. Zero Interval].
- *[Mandantory Zero Stop Time] can be set in the following sequence:
[Preset]→[Detailed Setting] →[Timing Setting]→[Mandantory Zero Stop Time].

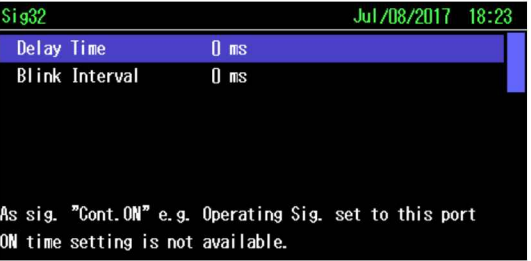
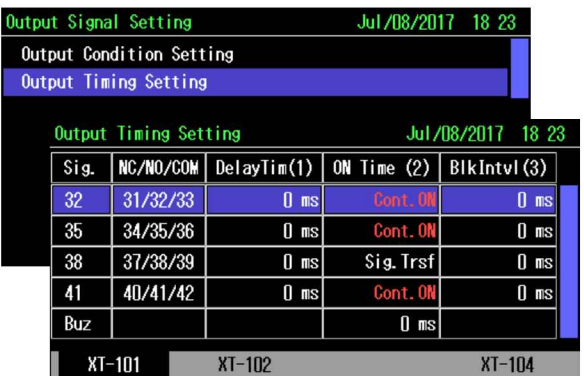
■ Output Signal Setting

This function sets the output signal setting.

1. Point the cursor to [Output Signal Setting], and press  .
 - [Output Signal Setting] screen will appear.
2. Press   key to select [Output Condition Setting] and press  .
 - [Output Condition Setting] screen will appear.



3. Point the cursor to [Output Timing Setting], and press  .
 - XT101 [Output Timing Setting] screen will appear.
4. Press   key to select signal and press  .
5. Point the cursor to [Delay Time], and press  .
 - Set the values screen will appear.



6. Set the values with the ten-key pad, press  .



5 Description of Maintenance Service Level

7. Press [Function 2], the setting screen for each setting value of XT-102 is displayed.

8. Press   key to select signal and press .

9. Point the cursor to [Delay Time], and press



• Set the values screen will appear.

10. Set the values with the ten-key pad, press .

11. Press [Function 4], XT104 [Output Timing Setting] screen will appear.

12. Press   key to select signal and press .

13. Point the cursor to [Delay Time], and press .

• Set the values screen will appear.

14. Set the values with the ten-key pad, press .

Output Timing Setting				Jul/08/2017 18:24	
Sig.	NC/NO/COM	DelayTim(1)	ON Time (2)	BlkIntvl(3)	
32	31/32/33	0 ms	Sig. Trsf	0 ms	
35	34/35/36	0 ms	Sig. Trsf	0 ms	
38	37/38/39	0 ms	Sig. Trsf	0 ms	
41	40/41/42	0 ms	Sig. Trsf	0 ms	
Buz			0 ms		
XT-101		XT-102	XT-104		

Sig32		Jul/08/2017 18:25	
Delay Time	0 ms		
ON Time	0 ms (Sig. Trsf)		
Blink Interval	0 ms		

Sig32		Jul/08/2017 18:24	
Delay Time	0 ms		
0			

Output Timing Setting				Jul/08/2017 18:25	
Sig.	NC/NO/COM	DelayTim(1)	ON Time (2)	BlkIntvl(3)	
32	31/32/33	0 ms	Sig. Trsf	0 ms	
35	34/35/36	0 ms	Sig. Trsf	0 ms	
38	37/38/39	0 ms	Sig. Trsf	0 ms	
41	40/41/42	0 ms	Sig. Trsf	0 ms	
Buz			0 ms		
XT-101		XT-102		XT-104	

Sig32		Jul/08/2017 18:25	
Delay Time	0 ms		
ON Time	0 ms (Sig. Trsf)		
Blink Interval	0 ms		

Sig32		Jul/08/2017 18:24	
Delay Time	0 ms		

■ **Mandatory Automatic Zero point**

This function sets the interval for the Mandatory Automatic Zero point.

- 1. Point the cursor to [Mand. Zero Interval], and press  .
 - Set the values screen will appear.
- 2. Set the values with the ten-key pad, press  .

Auto Zero Setting		Jul/08/2017 18 26
Auto Zero Weight	1.5 g	
Auto Zero Stable Wt.	0.4 g	
Auto Zero Adj. Average No.	32 pc	
Zero Check at Operation Start	Off	
Criteria of Wt.Fluc.	Off	
Mandantory Zero	Off	
Output Signal Setting		
Mand. Zero Interval	0 min	

Auto Zero Setting		0 minJul/08/2017 18 26
Mand. Zero Interval		
	0	

5.1.5 Weigh Unit Setting

- 1. Point the cursor to [Weigh Unit Setting], and press  .
 - [Weigh Unit Setting] menu will appear.
- 2. Point the cursor to the item you wish to set, and press  .(The Common setting items are displayed on the screen, however some of them cannot be changed their setting values.)

Setup Menu		Jul/08/2017 18:32
System Configuration		
System Diagnostics		
Auto Zero Setting		
Weigh Unit Setting		
Drive Unit Setting		
Communication Unit Setting		
Buzzer Setting		
Screen Capture Setting		

Weigher		GN-S015	Jul/08/2017 18 33
Applicable Country	USA		
Weigh Unit	On		
Maximum	1500.0 g		
AFV Adjustment			
Span Unit Count	265289728		
Zero AD Count	0		
Notch Frequency R	0.00 Hz		

Weigh Unit Setting		Jul/08/2017 18:34
Zero AD Count	0	
Notch Frequency R	0.00 Hz	
Notch Frequency M	0.00 Hz	
Edge modulus	0	
Weight	0	
Span Count	0	
Span Capacity	± 75.0 g	
Manual Zero Weight	± 150.0 g	

- 3. Press  .

"Weigh unit setting in process: Please wait." message will appear, then returns to [Weigh unit setting] screen.

Weigh Unit Setting		Jul/08/2017 18 34
Weigh Unit Setting		
Weigh unit setting in progress: Please wait.		

5 Description of Maintenance Service Level

5.1.5.1 AFV Setting (optional)

The AFV setting is initialized with the setting value of the standard specification when the product setting is configured. However, weight of optional unit must be entered for weigh conveyor with optional units (hand over plate, guide).

1. Point the cursor to [Weigh Unit Setting], and press .

• [Weigh Unit Setting] menu will appear.

Setup Menu		Jul/08/2017 18:32
System Configuration		
System Diagnostics		
Auto Zero Setting		
Weigh Unit Setting		
Drive Unit Setting		
Communication Unit Setting		
Buzzer Setting		
Screen Capture Setting		

2. Point the cursor to [AFV Adjustment], and press .

Weigh Unit Setting		Jul/08/2017 18:35
Weigher	GN-S015	
Applicable Country	USA	
Weigh Unit	On	
Maximum	1500.0 g	
AFV Adjustment		
Span Unit Count	265289728	
Zero AD Count	0	
Notch Frequency R	0.00 Hz	

3. Select "Off" and press .

AFV Adjustment		Jul/08/2017 18:36
AFV Select	Off	
Weigh Belt Dead Wt.	680 x10g	
Additional Dead Wt.	0 x10g	
AFV Coefficient	3003	
Dead Wt. Coefficient	2282	

4. Press   key to select "On" and press .

AFV Adjustment		
AFV Select	Off	Jul/08/2017 18:36
0	Off	
1	On	

AFV Adjustment		
AFV Select	Off	Jul/08/2017 18:36
0	Off	
1	On	

5. Point the cursor to [Additional Dead Wt.] and press .

AFV Adjustment		Jul/08/2017 18:37
AFV Select	On	
Weigh Belt Dead Wt.	680 x10g	
Additional Dead Wt.	0 x10g	
AFV Coefficient	3003	
Dead Wt. Coefficient	2282	
AFV Simulation		

6. Input the weight of the transfer plate and guide etc., and press .

AFV Adjustment		Jul/08/2017 18:38
Additional Dead Wt.	0 x10g	
	47	

7. Press .

AFV Adjustment		Jul/08/2017 18:40
AFV Select	On	
Weigh Belt Dead Wt.	680 x10g	
Additional Dead Wt.	47 x10g	
AFV Coefficient	3210	
Dead Wt. Coefficient	2134	
AFV Simulation		

- [Weigh Unit Setting] menu will appear.

Weigh Unit Setting		Jul/08/2017 18:35
Weigher	GN-S015	
Applicable Country	USA	
Weigh Unit	On	
Maximum	1500.0 g	
AFV Adjustment		
Span Unit Count	265289728	
Zero AD Count	0	
Notch Frequency R	0.00 Hz	

8. "Weigh unit setting in process:Please wait. " message will appear, then returns to [Weigh unit setting] screen.

Weigh Unit Setting		Jul/08/2017 18:34
Weigh Unit Setting		
Weigh unit setting in progress: Please wait.		

Weigh Unit Setting		Jul/08/2017 18:35
Weigher	GN-S015	
Applicable Country	USA	
Weigh Unit	On	
Maximum	1500.0 g	
AFV Adjustment		
Span Unit Count	265289728	
Zero AD Count	0	
Notch Frequency R	0.00 Hz	



Refer to AFV Mounting of "4.5.4.1 Mounting AFV unit".

5.1.6 Drive Unit Setting

1. Point the cursor to [Drive Unit Setting], and press



- [Drive Unit Setting] menu will appear.

Setup Menu	Jul/08/2017 18:43
System Configuration	
System Diagnostics	
Auto Zero Setting	
Weigh Unit Setting	
Drive Unit Setting	
Communication Unit Setting	
Buzzer Setting	
Screen Capture Setting	

Drive Unit Setting	Jul/08/2017 18:43
IF Conveyor rotation	CW
IFC IP Shaft-Gear ratio	14
IFC OP Shaft-Gear ratio	22
IFC D-Roller Dia.	24.00
Infeed Conv. Gear Head	1/1
Infeed Conv. Min Speed	5 m/min
Infeed Conv. Max Speed	120 m/min
Weigh Conveyor Length	450 mm

Drive Unit Setting	Jul/08/2017 18:45
WT Conveyor rotation	CW
WTC IP Shaft-Gear ratio	14
WTC OP Shaft-Gear ratio	22
WTC D-Roller Dia.	24.00
Weighing Conv. Gear Head	1/1
WT Conveyor Min Speed	5 m/min
WT Conveyor Max Speed	120 m/min
Drive Unit Setting(REJ)	

■ IF Conveyor rotation

1. Point the cursor to [IF Conveyor rotation], and



- The selection screen for the rotation direction is displayed.

2. Press



key to select "CCW" and



press

Drive Unit Setting	Jul/08/2017 18:43
IF Conveyor rotation	CW
IFC IP Shaft-Gear ratio	14
IFC OP Shaft-Gear ratio	22
IFC D-Roller Dia.	24.00
Infeed Conv. Gear Head	1/1
Infeed Conv. Min Speed	5 m/min
Infeed Conv. Max Speed	120 m/min
Weigh Conveyor Length	450 mm

Drive Unit Setting	IF Conveyor rotation	CW	Jul/08/2017 18:44
0	CW		
1	CCW		

Drive Unit Setting	IF Conveyor rotation	CW	Jul/08/2017 18:44
0	CW		
1	CCW		

3. The rotational direction of the infeed conveyor was changed from CW to CCW.

Drive Unit Setting	Jul/08/2017 18:43
IF Conveyor rotation	CCW
IFC IP Shaft-Gear ratio	14
IFC OP Shaft-Gear ratio	22
IFC D-Roller Dia.	24.00
Infeed Conv. Gear Head	1/1
Infeed Conv. Min Speed	5 m/min
Infeed Conv. Max Speed	120 m/min
Weigh Conveyor Length	450 mm

■ Weigh Conveyor Length

1. Point the cursor to [Weigh Conveyor Length], and press .

• [Weigh Conveyor Length] menu will appear.

2. To change the weigh conveyor length, enter the value by using the ten-keypad, and press the



• The screen returns to Drive Unit Setting.

Drive Unit Setting		Jul/08/2017 18 50
IF Conveyor rotation	CW	
IFC IP Shaft-Gear ratio	14	
IFC OP Shaft-Gear ratio	22	
IFC D-Roller Dia.	24.00	
Infeed Conv. Gear Head	1/1	
Infeed Conv. Min Speed	5 m/min	
Infeed Conv. Max Speed	120 m/min	
Weigh Conveyor Length	450 mm	

Drive Unit Setting		Jul/08/2017 18 50
Weigh Conveyor Length	450 mm	
	450	

Drive Unit Setting		Jul/08/2017 18 50
IF Conveyor rotation	CW	
IFC IP Shaft-Gear ratio	14	
IFC OP Shaft-Gear ratio	22	
IFC D-Roller Dia.	24.00	
Infeed Conv. Gear Head	1/1	
Infeed Conv. Min Speed	5 m/min	
Infeed Conv. Max Speed	120 m/min	
Weigh Conveyor Length	450 mm	

■ WT Conveyor rotation

1. Point the cursor to [WT Conveyor rotation], and press .

• [WT Conveyor rotation] menu will appear.

2. Press   key to select "CCW" and press .

Drive Unit Setting		Jul/08/2017 18 51
WT Conveyor rotation	CW	
WTC IP Shaft-Gear ratio	14	
WTC OP Shaft-Gear ratio	22	
WTC D-Roller Dia.	24.00	
Weighing Conv. Gear Head	1/1	
WT Conveyor Min Speed	5 m/min	
WT Conveyor Max Speed	120 m/min	
Drive Unit Setting(REJ)		

Drive Unit Setting		Jul/08/2017 18:52
WT Conveyor rotation	CW	
0	CW	
1	CCW	

Drive Unit Setting		Jul/08/2017 18 52
WT Conveyor rotation	CW	
0	CW	
1	CCW	

Drive Unit Setting		Jul/08/2017 18 51
WT Conveyor rotation	CCW	
WTC IP Shaft-Gear ratio	14	
WTC OP Shaft-Gear ratio	22	
WTC D-Roller Dia.	24.00	
Weighing Conv. Gear Head	1/1	
WT Conveyor Min Speed	5 m/min	
WT Conveyor Max Speed	120 m/min	
Drive Unit Setting(REJ)		



[Table for initial data of each model]

SS Standard									
015-*/SS-I									
Weigh conveyor type	11	12	13	14	22	23	24	33	34
Unit length	605	605	705	805	605	705	805	705	805
Weigh conveyor length	150	250	350	450	250	350	450	350	450
Infeed conveyor length	450	350	350	350	350	350	350	350	350
Conveyor gap	5	5	5	5	5	5	5	5	5
Weigh conveyor roller diameter	18	18	18	18	24	24	24	24	24
Weigh conveyor pulley gear teeth number	14	14	14	14	22	22	22	22	22
Weigh motor pulley gear teeth number	14	14	14	14	14	14	14	14	14
Infeed conveyor roller diameter	18	18	18	18	24	24	24	24	24
Infeed conveyor pulley gear teeth number	14	14	14	14	22	22	22	22	22
Infeed motor pulley gear teeth number	14	14	14	14	14	14	14	14	14

SS Standard									
060-*/SS-I									
Weigh conveyor type	22	23	24	33	34	35	36	44	45
Unit length	605	705	805	705	905	1005	1205	900	1000
Weigh conveyor length	250	350	450	350	450	550	650	445	545
Infeed conveyor length	350	350	350	350	450	450	550	445	545
Conveyor gap	5	5	5	5	5	5	5	10	10
Weigh conv. roller dia.	24	24	24	24	24	24	24	33	33
Weigh conv. pulley gear teeth No.	22	22	22	22	22	22	22	30	30
Weigh motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14
Infeed conv. roller dia.	24	24	24	24	24	24	24	33	33
Infeed conv. pulley gear teeth No.	22	22	22	22	22	22	22	30	30
Infeed motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14

CR Standard									
015-*/CR-I									
Weigh conveyor type	12	13	14	22	23	24	33	34	
Unit length	600	700	800	600	700	800	700	800	
Weigh conveyor length	245	345	445	245	345	445	345	445	
Infeed conveyor length	345	345	345	345	345	345	345	345	
Conveyor gap	10	10	10	10	10	10	10	10	
Weigh conveyor roller diameter	33	33	33	33	33	33	33	33	
Weigh conveyor pulley gear teeth number	30	30	30	30	30	30	30	30	
Weigh motor pulley gear teeth number	14	14	14	14	14	14	14	14	
Infeed conveyor roller diameter	33	33	33	33	33	33	33	33	
Infeed conveyor pulley gear teeth number	30	30	30	30	30	30	30	30	
Infeed motor pulley gear teeth number	14	14	14	14	14	14	14	14	



[Table for initial data of each model]

CR Standard										
060-*/CR-I										
Weigh conveyor type	22	23	24	33	34	35	36	44	45	46
Unit length	600	700	800	700	900	1000	1200	900	1000	1200
Weigh conveyor length	245	345	445	345	445	545	645	445	545	645
Infeed conveyor length	345	345	345	345	445	445	545	445	445	545
Conveyor gap	10	10	10	10	10	10	10	10	10	10
Weigh conv. roller dia.	33	33	33	33	33	33	33	33	33	33
Weigh conv. pulley gear teeth No.	30	30	30	30	30	30	30	30	30	30
Weigh motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14	14
Infeed conv. roller dia.	33	33	33	33	33	33	33	33	33	33
Infeed conv. pulley gear teeth No.	30	30	30	30	30	30	30	30	30	30
Infeed motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14	14

SS w/Metal detector										
015-*/SS-S##										
Weigh conveyor type		12	13	14	22	23	24	33	34	
Unit length		945	945	1045	945	945	1045	1200	1300	
*At coil height of 180;		1045	1045	1145	1045	1045	1145	1200	1300	
Weigh conveyor length		250	350	450	250	350	450	350	450	
Infeed conveyor length		690	590	590	690	590	590	845	845	
*At coil height of 180;		790	690	690	790	690	690	845	845	
Conveyor gap		5	5	5	5	5	5	5	5	
Weigh conveyor roller diameter		18	18	18	24	24	24	24	24	
Weigh conveyor pulley gear teeth number		14	14	14	22	22	22	22	22	
Weigh motor pulley gear teeth number		14	14	14	14	14	14	14	14	
Infeed conveyor roller diameter		33	33	33	33	33	33	33	33	
Infeed conveyor pulley gear teeth number		30	30	30	30	30	30	30	30	
Infeed motor pulley gear teeth number		14	14	14	14	14	14	14	14	

SS w/Metal detector										
060-*/SS-S##										
Weigh conveyor type	22	23	24	33	34	35	36	44	45	46
Unit length	945	945	1045	1200	1300	1400	1500	1300	1400	1500
*At coil height of 180;	1045	1045	1145	1200	1300	1400	1500	1300	1400	1500
Weigh conveyor length	250	350	450	350	450	550	650	445	545	645
Infeed conveyor length	690	590	590	845	845	845	845	845	845	845
*At coil height of 180;	790	690	690	845	845	845	845	845	845	845
Conveyor gap	5	5	5	5	5	5	5	10	10	10
Weigh conv. roller diameter	24	24	24	24	24	24	24	33	33	33
Weigh conv. pulley gear teeth No.	22	22	22	22	22	22	22	30	30	30
Weigh motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14	14
Infeed conv. roller diameter	33	33	33	33	33	33	33	33	33	33
Infeed conv. pulley gear teeth No.	30	30	30	30	30	30	30	30	30	30
Infeed motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14	14



[Table for initial data of each model]

CR w/Metal detector								
015-*/CR-S##								
Weigh conveyor type	12	13	14	22	23	24	33	34
Unit length	945	945	1045	945	945	1045	1200	1300
*At coil height of 180;	1045	1045	1145	1045	1045	1145	1200	1300
Weigh conveyor length	245	345	445	245	345	445	345	445
Infeed conveyor length	690	590	590	690	590	590	845	845
*At coil height of 180;	790	690	690	790	690	690	845	845
Conveyor gap	10	10	10	10	10	10	10	10
Weigh conveyor roller diameter	33	33	33	33	33	33	33	33
Weigh conveyor pulley gear teeth number	30	30	30	30	30	30	30	30
Weigh motor pulley gear teeth number	14	14	14	14	14	14	14	14
Infeed conveyor roller diameter	33	33	33	33	33	33	33	33
Infeed conveyor pulley gear teeth number	30	30	30	30	30	30	30	30
Infeed motor pulley gear teeth number	14	14	14	14	14	14	14	14

CR w/Metal detector										
060-*/CR-S##										
Weigh conveyor type	22	23	24	33	34	35	36	44	45	46
Unit length	945	945	1045	1200	1300	1400	1500	1300	1400	1500
*At coil height of 180;	1045	1045	1145	1200	1300	1400	1500	1300	1400	1500
Weigh conveyor length	245	345	445	345	445	545	645	445	545	645
Infeed conveyor length	690	590	590	845	845	845	845	845	845	845
*At coil height of 180;	790	690	690	845	845	845	845	845	845	845
Conveyor gap	10	10	10	10	10	10	10	10	10	10
Weigh conv. roller dia.	33	33	33	33	33	33	33	33	33	33
Weigh conv. pulley gear teeth No.	30	30	30	30	30	30	30	30	30	30
Weigh motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14	14
Infeed conv. roller dia.	33	33	33	33	33	33	33	33	33	33
Infeed conv. pulley gear teeth No.	30	30	30	30	30	30	30	30	30	30
Infeed motor pulley gear teeth No.	14	14	14	14	14	14	14	14	14	14

■ Drive unit setting (rejector)

Drive Unit Setting window enables to change the drive unit gear ratio of rejector.

1. Point the cursor to [Rejector Parameter Setup], and press .

2. Point the cursor to the parameter to change, and press .

• Input the value to change with the ten-key pad.

3. Press .

Drive Unit Setting		Mar/02/2016 9:40
IFC IP Shaft-Gear ratio	14	
IFC OP Shaft-Gear ratio	22	
IFC Drv. Rllr. D	24.00	
WTC IP Shaft-Gear ratio	14	
WTC OP Shaft-Gear ratio	22	
WTC D-Roller Dia.	24.00	
IFC Fixed Speed	20 m/min	
Rejector Parameter Setup		

Reject		Mar/02/2016 9:40
Reject Conveyor Length	550 mm	
RJC IP Shaft-Gear ratio	14	
RJC OP Shaft-Gear ratio	22	
RJC D-Roller Dia.	24.00	

Reject		Mar/02/2016 9:40
Reject Conveyor Length	550 mm	
550		

■ Change the reject conveyor rotation

1. Point the cursor to [Reject Conveyor Rotation], and press .

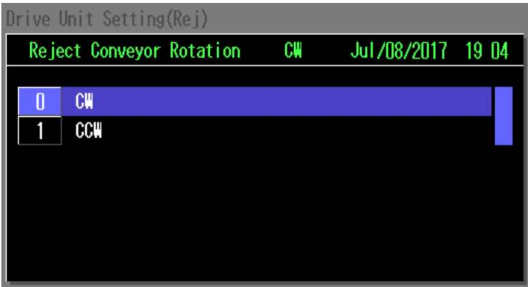
• The selection screen for the rotation direction is displayed.

Drive Unit Setting(Rej)		Jul/08/2017 19 04
Reject Conveyor Length	550 mm	
Reject Conveyor Rotation	CW	
RJC IP Shaft-Gear ratio	14	
RJC OP Shaft-Gear ratio	22	
RJC D-Roller Dia.	24.00	
Rej. Speed Reduce Ratio	1/1	
Reject Conv. Min. Spd.	5 m/min	
Reject Conv. Max. Spd.	120 m/min	

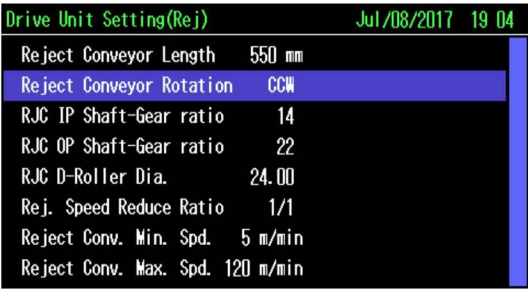
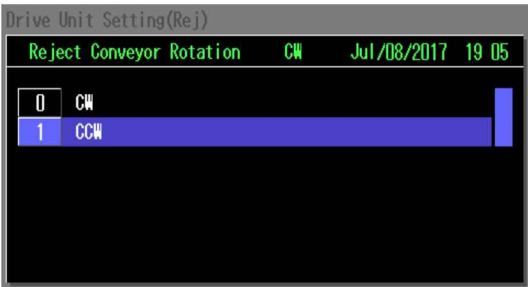
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5 Description of Maintenance Service Level

2. Press   key to select "CCW" and press .

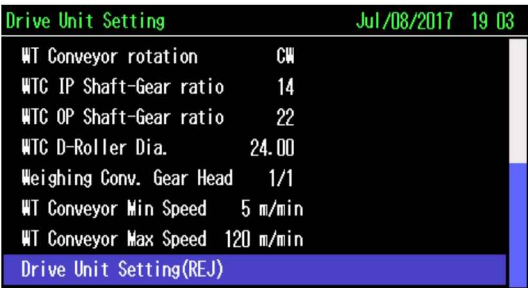


- The rotational direction of the reject conveyor was changed from "CW" to "CCW".



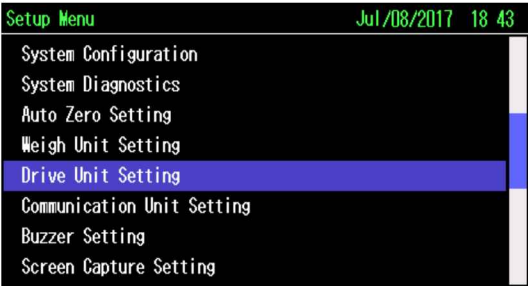
3. Press .

- [Drive Unit Setting (REJ)] menu will appear.



4. Press .

- [Drive Unit Setting] menu will appear.



5.1.7 Communication Unit Setting (optional)

Performs the setting for communication unit.



This section explains how to set the communication unit for standard software.
For detailed operation of the Optional Settings, refer to "7.5 Optional Settings".

1. Point the cursor to [Communication Unit Setting], and press .
 - [Communication Unit Setting] menu will appear.
2. Point the cursor to [COM0], and press .
 - Communication Unit Setting COM0 menu will appear.

NOTE The option [COM0] in Communication Unit Setting is the port specific to the metal detection, and is not settable (browse only).

3. Press key return to [Communication Unit Setting] screen.

Communication Unit Setting Jul/08/2017 20:48	
COM0	Non
COM1	Non
COM2	Non
Ethernet	

- "Non" is displayed to mean there is no communication target.
4. Press key to select "COM1" and press .
 5. In case of setting the communication target, point the cursor to "Target", and press .

Setup Menu Jul/08/2017 20:47	
System Configuration	
System Diagnostics	
Auto Zero Setting	
Weigh Unit Setting	
Drive Unit Setting	
Communication Unit Setting	
Buzzer Setting	
Screen Capture Setting	

Communication Unit Setting Jul/08/2017 20:48	
COM0	Non
COM1	Non
COM2	Non
Ethernet	

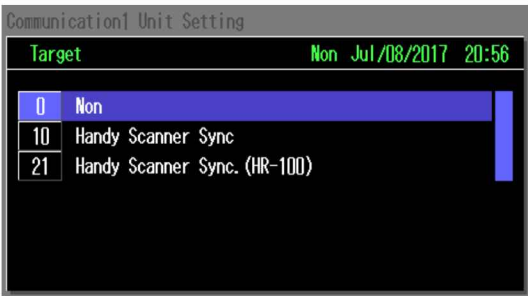
Communication[0] Unit Setting Jul/08/2017 20:49	
Target	Non
Speed	115200 bps
Data Bit Length	8 bit
Stop Bit Length	1 bit
Parity	Off

Communication Unit Setting Jul/08/2017 20:50	
COM0	Non
COM1	Non
COM2	Non
Ethernet	

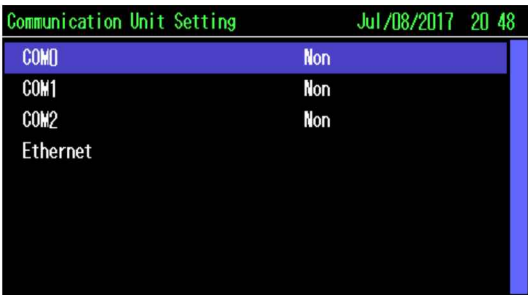
Communication[1] Unit Setting Jul/08/2017 20:50	
Target	Non
Speed	9600 bps
Data Bit Length	8 bit
Stop Bit Length	1 bit
Parity	Off

5 Description of Maintenance Service Level

6. Point the cursor to the communication target by the   key, and press the  key.



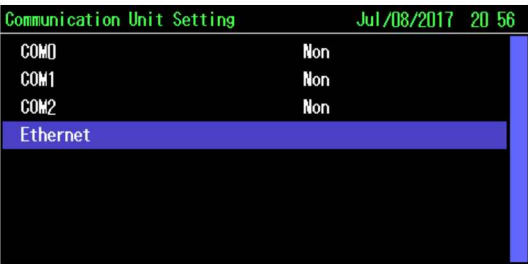
7. Press  two times to return to [Communication Unit Setting] screen.



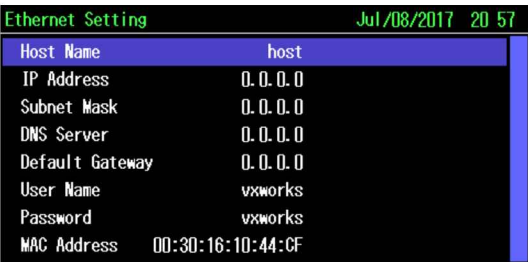
■ Check the Ethernet Setting

1. Press   key to select "Ethernet" and press .

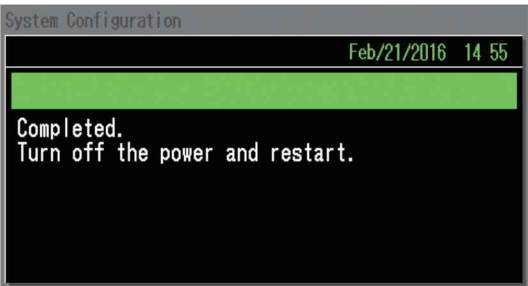
- [Ethernet] menu will appear.



2. The setting value as [IP Address] can be checked or changed.



3. Press  two times to return to Setup Menu screen.



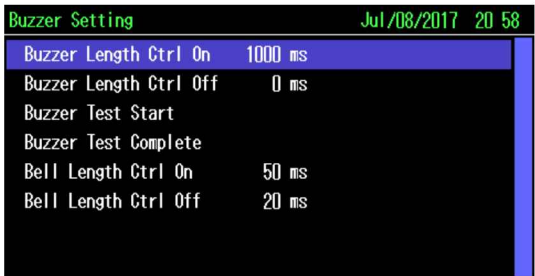
NOTE If the setting the communication target has been changed, "Completed. Turn off the power and restart." message will appear. Follow the instruction on the screen.
The communication setting is updated.

5.1.8 Buzzer Setting

1. Point the cursor to [Buzzer Setting], and press



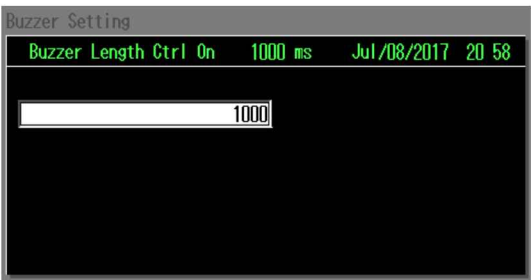
• [Buzzer Setting] menu will appear.



2. Point the cursor to the item you wish to set, and



Set the values with the ten-key pad, press



5.1.9 Screen Capture

Setting Screen Capture to "On" enables to save the screen currently displayed into the USB memory.

■ Screen capture

1. Create a folder for capture (named as "capture") on the PC, and save the folder into the USB memory.

2. Point the cursor to [Screen Capture Setting], and



3. Point the cursor to [Capture Screen by F3 Key] and press

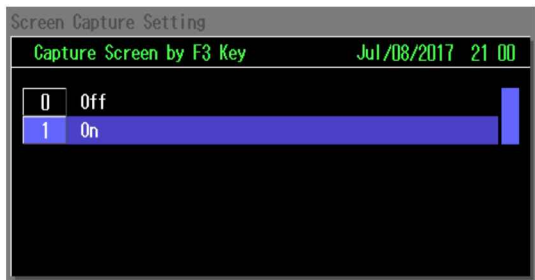
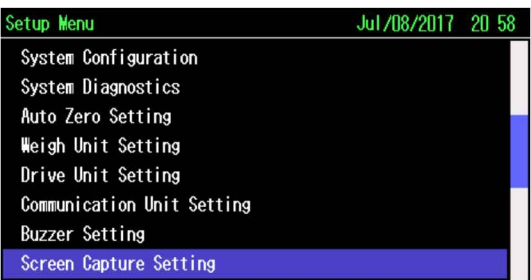


• [Screen Capture] On or Off menu select will appear.

4. Press



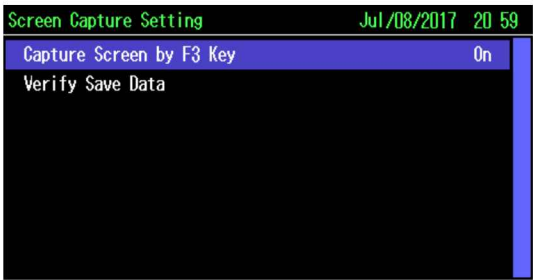
key to select "On", and press



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5 Description of Maintenance Service Level

- [Capture Screen by F3 Key] menu will appear.



5. Press  key return to [Setup Menu] screen.

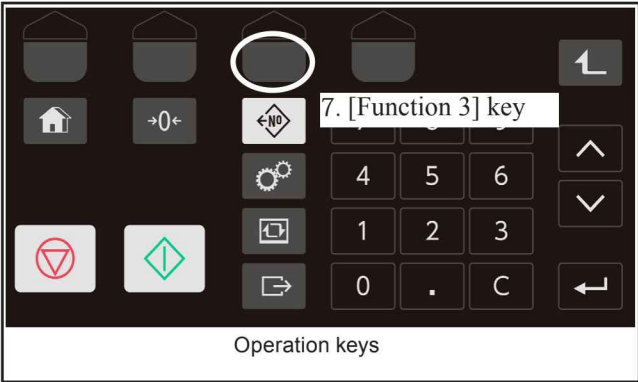


6. Display the screen to capture.

7. Press [Function 3] key.



8. The screen "xxx. jpg." is saved into the "capture" folder in the USB memory.



NOTE The screen of [Filter Simulation] cannot be captured even when the [Screen Capture] is set to "On".

■ Verify save data

1. Point the cursor to [Screen Capture Setting], and press .

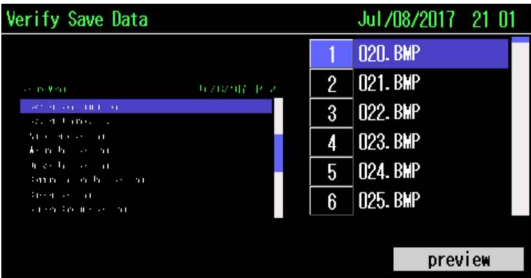
- [Screen Capture Setting] screen will appear.



2. Press  key to select [Verify Save Data], and press .

- [Screen Capture Setting] screen will appear.

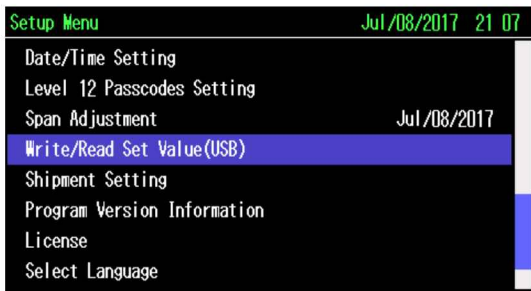
3. Press  two times to return to Setup Menu screen.



5.1.10 Setting Value Save/Recovery

1. Point the cursor to [Write/Read set Value (USB)], and press .

- [Write/Read set Value (USB)] menu will appear.



2. Point the cursor to the item you wish to set, and press  to display the confirmation screen. Press  to complete the setting.

[Setting menu]

```

Write/Read Set Value(USB)          Jul/08/2017 21:08
Write Common Setting (SETUP_GNC)
Write Preset Setting (PRESET_GNC)
Preset Setting CSV Output(Preset)

Read Common Setting (SETUP_GNC)
Read Preset Setting (PRESET_GNC)
Preset Setting CSV Input(Preset)

```

[Confirmation message]

```

Write/Read Set Value(USB)          Jul/08/2017 21 09
Write the setting to the USB memory.
The setting is written to the USB memory.
Press [Exit] key to cancel.
Press [Enter] key to execute.

```

```

Write/Read Set Value(USB)          Jul/08/2017 21:10
Write Common Setting (SETUP_GNC)
Write Preset Setting (PRESET_GNC)
Preset Setting CSV Output(Preset)

Read Common Setting (SETUP_GNC)
Read Preset Setting (PRESET_GNC)
Preset Setting CSV Input(Preset)

```

```

Write/Read Set Value(USB)          Jul/08/2017 21 09
Write the setting to the USB memory.
The setting is written to the USB memory.
Press [Exit] key to cancel.
Press [Enter] key to execute.

```

```

Write/Read Set Value(USB)          Jul/08/2017 21 12
Write Common Setting (SETUP_GNC)
Write Preset Setting (PRESET_GNC)
Preset Setting CSV Output(Preset)

Read Common Setting (SETUP_GNC)
Read Preset Setting (PRESET_GNC)
Preset Setting CSV Input(Preset)

```

```

Write/Read Set Value(USB)          Jul/08/2017 21 12
Write the setting to the USB memory.
Preset setting is written to the USB memory.
Press [Exit] key to cancel.
Press [Enter] key to execute.

```

```

Write/Read Set Value(USB)          Jul/08/2017 21:19
Write Common Setting (SETUP_GNC)
Write Preset Setting (PRESET_GNC)
Preset Setting CSV Output(Preset)

Read Common Setting (SETUP_GNC)
Read Preset Setting (PRESET_GNC)
Preset Setting CSV Input(Preset)

```

```

Write/Read Set Value(USB)          Jul/26/2017 16:55
No Files in USB.
Press [Exit] key.
10505-00

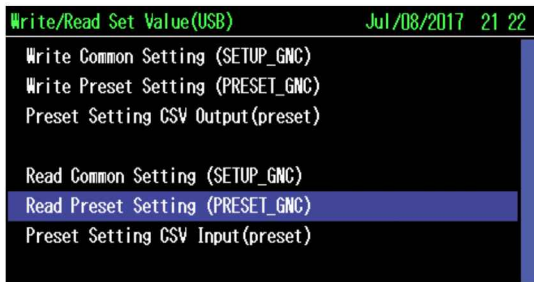
```

```

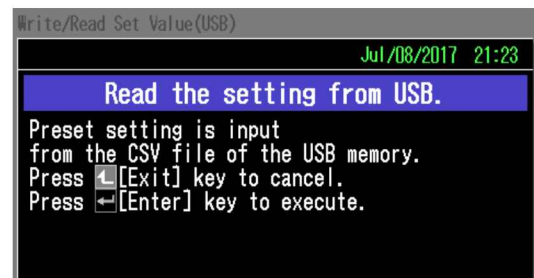
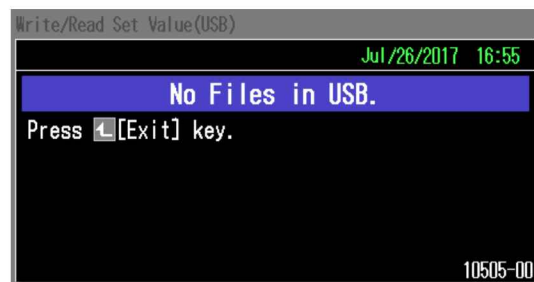
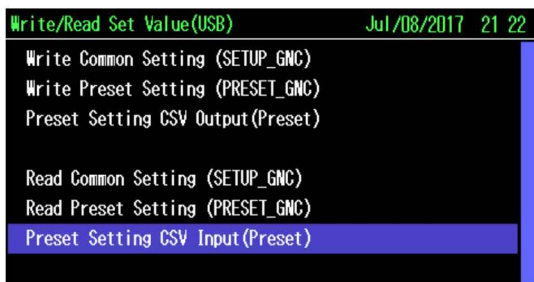
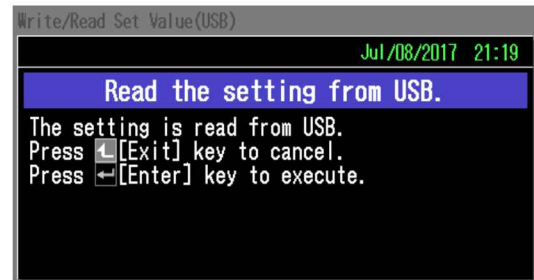
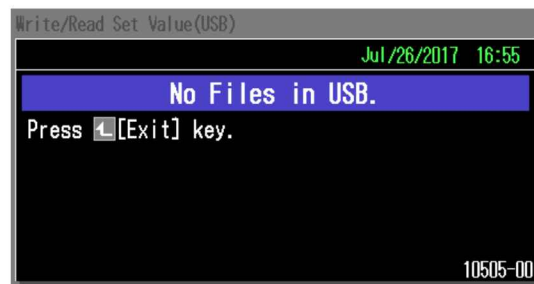
Write/Read Set Value(USB)          Jul/08/2017 21:19
Read the setting from USB.
The setting is read from USB.
Press [Exit] key to cancel.
Press [Enter] key to execute.

```

[Setting menu]



[Confirmation message]

**NOTE**

Before starting the system initialization, be sure to save the common settings and pre-set settings to the USB memory, by "Setting Value Save/Recovery".

When you use plural check weighers which model and specification are the same, the settings can be copied from the USB memory by using the Common Setting function.

5.1.11 Shipment Setting

In the shipment setting, Common Setting and Preset Setting according to the specification are initially saved in the MCU board memory incorporated in the checkweigher.

NOTE

- When Common Setting and/or Preset Setting are changed after the delivery, be sure to save the change USB memory by "Save the Shipment setting".
- Refer to section "5.1.10 Setting Value Save/Recovery" and save the backup data for Common settings and Preset settings.

TIP

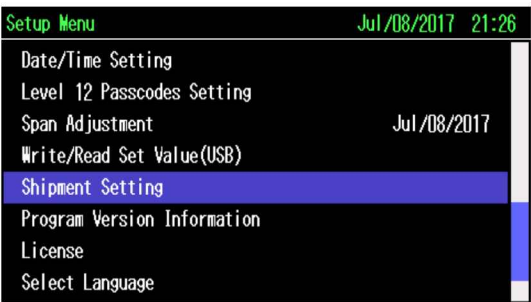
Even in the cases where the wrong settings are given or when the system initialization is mistakenly executed, you can return to the settings for which the last "Save the Shipment setting" was executed. Execute "Read Common Setting" and "Read Preset Setting" of "Shipment setting".

1.

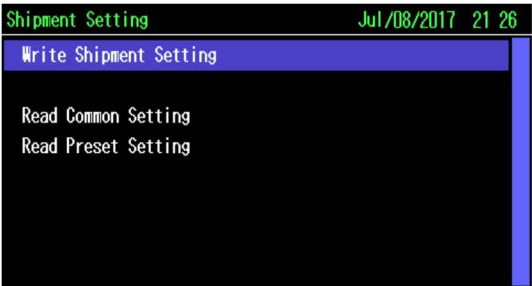
Point the cursor to [Shipment setting], and press .

- [Shipment setting] menu will appear.
2.

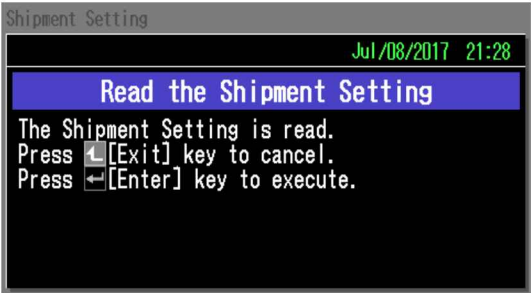
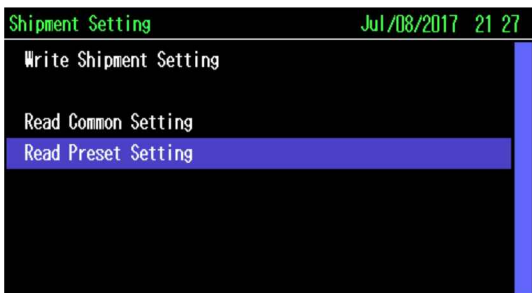
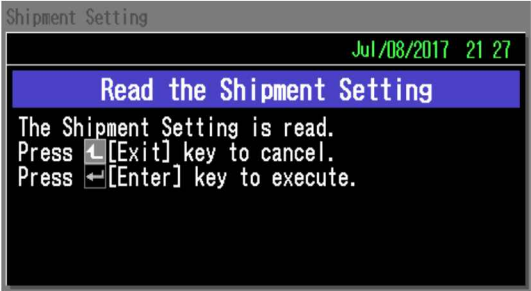
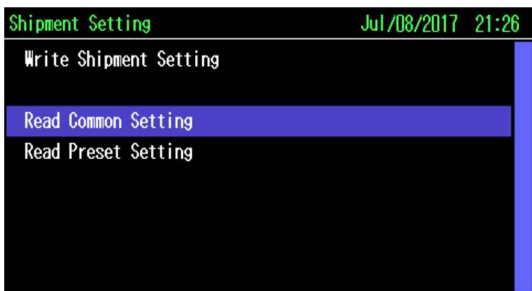
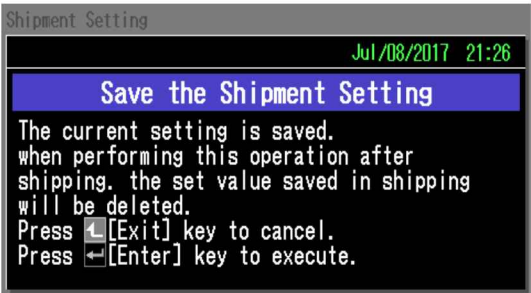
Point the cursor to the item you wish to set, and press  key to display the confirmation screen. Press  key to complete the setting.



[Setting menu]

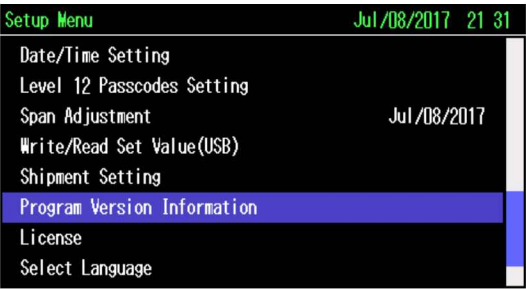


[Confirmation message]

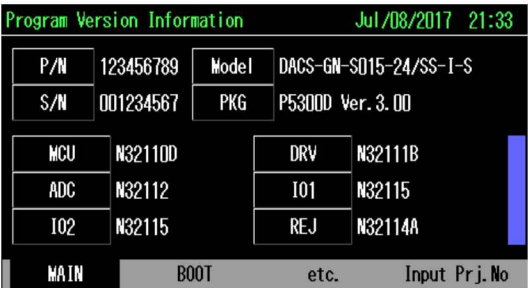


5.1.12 Program Version Information

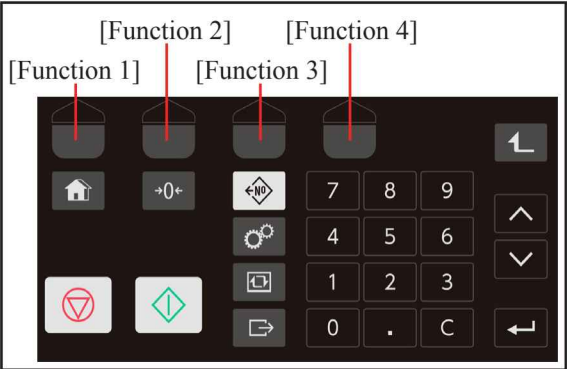
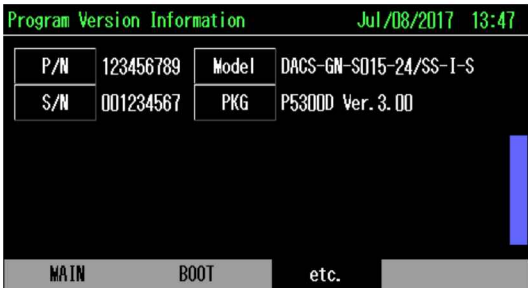
1. Point the cursor to [Program Version Information],
and press .
- [Program Version Information] screen
will appear, and you can check the version
information of the program installed in this
machine.
2.  key can switch the screen.



Press [Function 1] key.



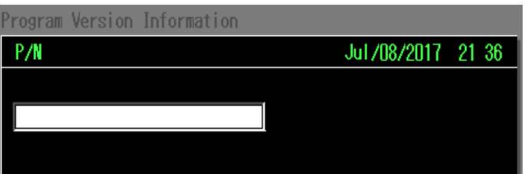
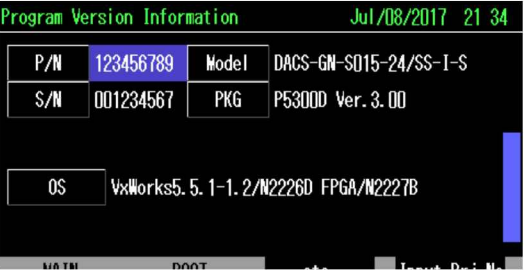
Press [Function 3] key.



Press [Function 2] key.



Press [Function 4] key.



3. Press [Function 4] key.
- The project No. (P/N) is selected.

Pressing the  key in this state can change the project No.

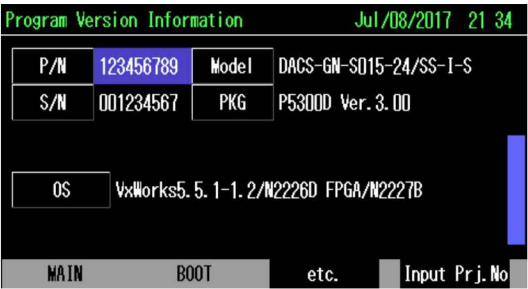


- [F1] key: displays the No. of main program.
[F2] key: displays the No. of boot program.
[F3] key: displays the No. of the program of OS or others.
[F4] key: selects the project No. Pressing the  key can change the project No.

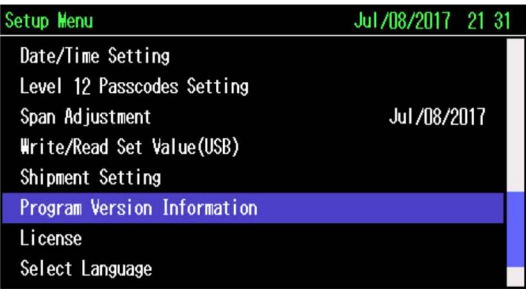
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5 Description of Maintenance Service Level

4. Set the P/N with the ten-key pad, press 



- 5 Press  to return to [Setup Menu] screen.



For the models which do not include I/O device, no program No. appears at [REJ], [IO-1], [IO-2] on the [Program Ver.] screen.

5.1.13 License

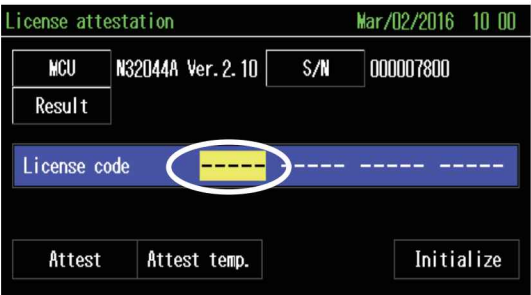
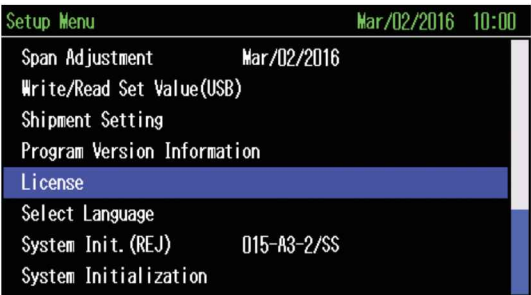
Optional devices are available only after purchasing and attesting the license.
Here describes the procedure for the attestation, temporal attestation and initialization of the license.



For the detailed procedure of licence attestation, temporary attestation, and initialization, refer to "7.3 License Confirmation".

REFERENCE

1. Point the cursor to [License] and press  .
- The License menu will appear.
2. Point the cursor to [License attestation] and press  .
- The License attestation menu will appear.



3. Point the cursor to [Show license] and press



• The [Show license] menu will appear.

Show License Jul/08/2017 21:42	
i-STATION LINK	Non
Wei Output	Non
Floating Ref. Weight	Non
Reject Mode	Non
Preset Exchange	Non
Host Interface	Non
Proper Count	Non
FBC Control	Non

License Jul/08/2017 21:40	
License Attestation	
Show License	
License Code History	

4. Point the cursor to [license code history] and



• [license code history] menu will appear.

License Code History Jul/08/2017 21:45	
Comp.	-----

Show License Jul/08/2017 21:42	
Floating Ref. Weight	Non
Reject Mode	Non
Preset Exchange	Non
Host Interface	Non
Proper Count	Non
FBC Control	Non
Rank Select	Non
i-FORT	Non

License Jul/08/2017 21:44	
License Attestation	
Show License	
License Code History	

5.1.14 Select Language

This function sets the language for the screen display.

1. Point the cursor to [Select Language], and press



• [Select Language] menu will appear.

Setup Menu Jul/08/2017 21:46	
Date/Time Setting	
Level 12 Passcodes Setting	
Span Adjustment	Jul/08/2017
Write/Read Set Value(USB)	
Shipment Setting	
Program Version Information	
License	
Select Language	

2. Press  key to select language, and press



• The screen is displayed in the selected language.

Select Language Jul/08/2017 21:47	
SELECT LANGUAGE.	
0	日本語 (Japanese)
1	English (English)
2	Francais (French)
3	Espanol (Spanish)
4	Portugues (Portuguese)



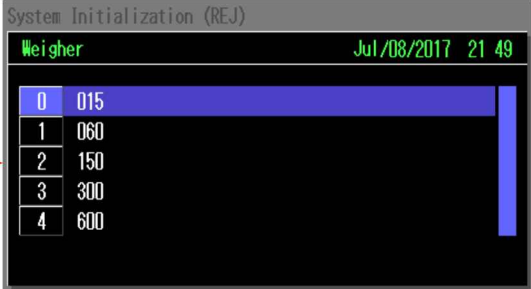
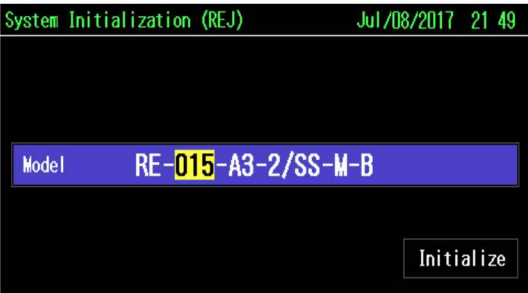
As of April 2017, the languages available are Japanese, English, French, Spanish, and Portuguese.

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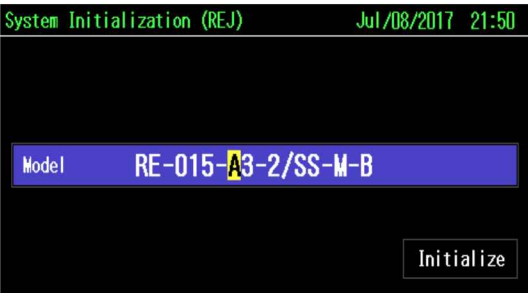
5.1.15 System Initialization (Rejector)

Select the model name of rejector that interlocks with the machine.

- 1. Point the cursor to [System Initialization (REJ)], and press .
- [System Initialization (REJ)] menu will appear.
- 2. Press .
- Selection screen for Weigher appears.



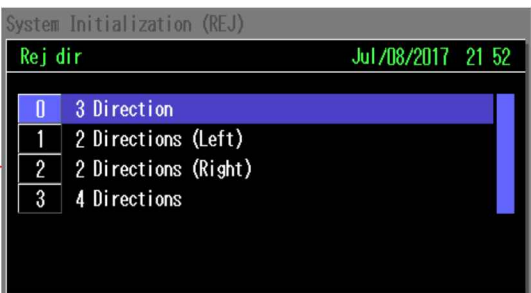
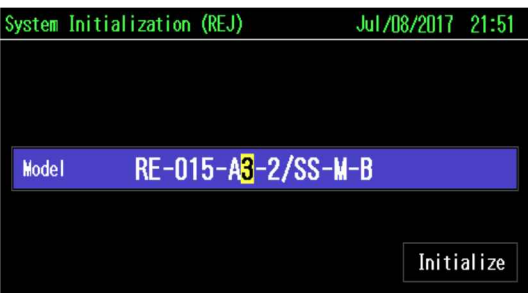
- 3. Press . Press  key to select of weigher, and press .



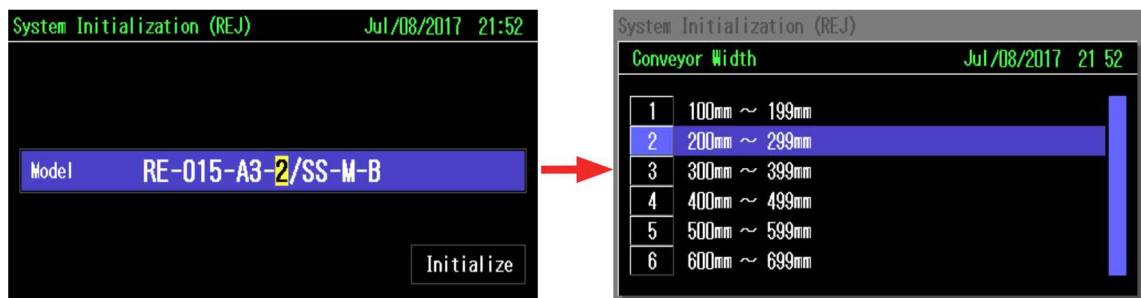
NOTE Be sure to select the correct rejector model; otherwise, a malfunction may occur at interlocking operations.



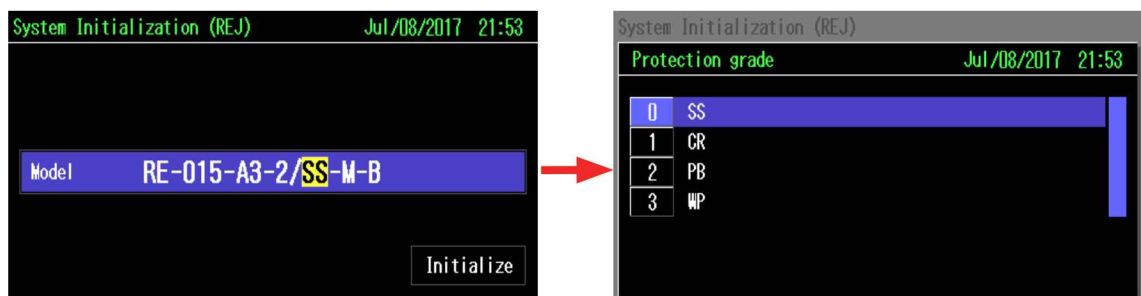
- 4. Press . Press  key to select of Rejection direction, and press .



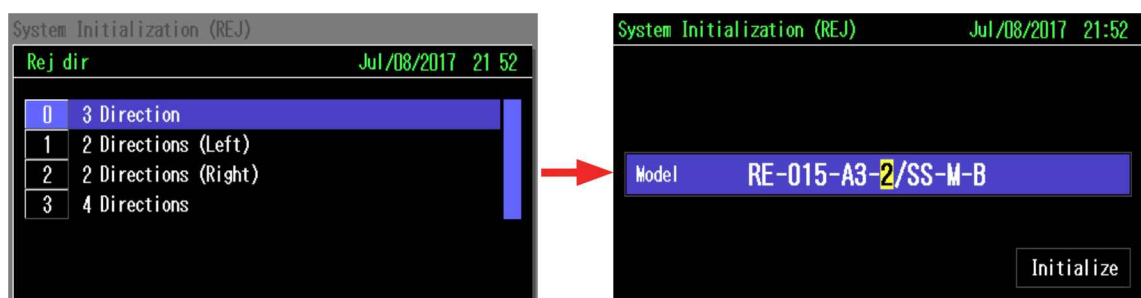
5. Press . Press key to select of Conveyor Width, and press .



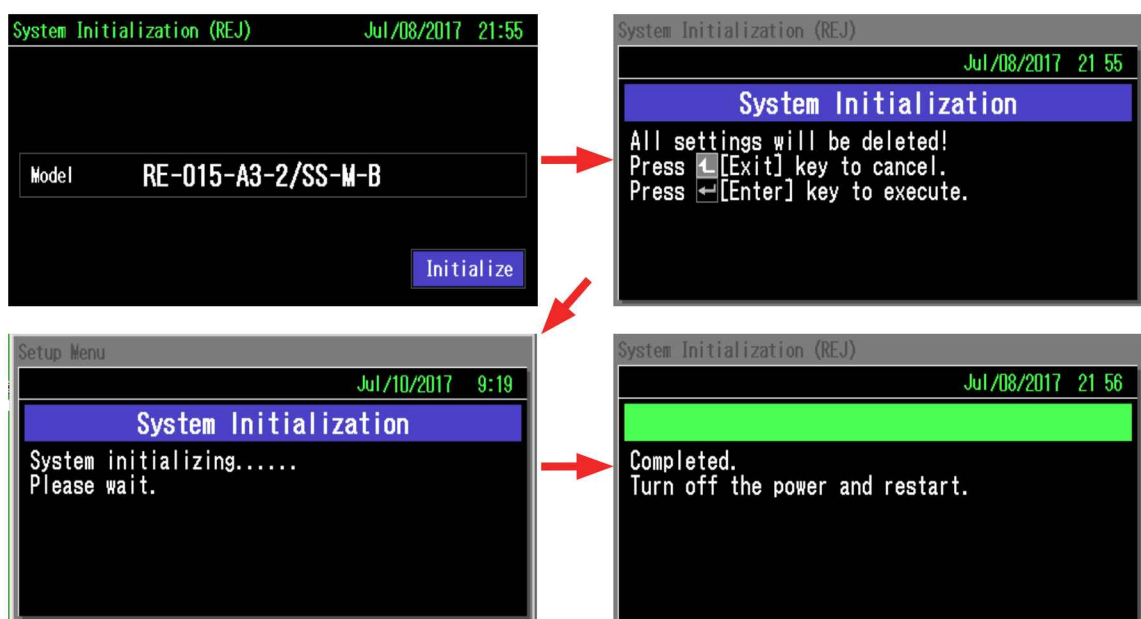
6. Press . Press key to select of Protection grade, and press .



7. Press . Press key to select of Drive system, and press .



8. Press . Press key to select of System Initialization, and press .



9. The message "Completed. Turn off the power and restart." is displayed. Follow the prompt on the screen.
(The setting data is updated in this occasion.)

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5.1.16 System Initialization

System Initialization, when executed, will lose the common setting parameters/preset setting parameters that have been previously memorized in the checkweigher, and return such setting to the default status. Note that the detailed weigh setting data are not initialized by the Initialization function at the Level 3.

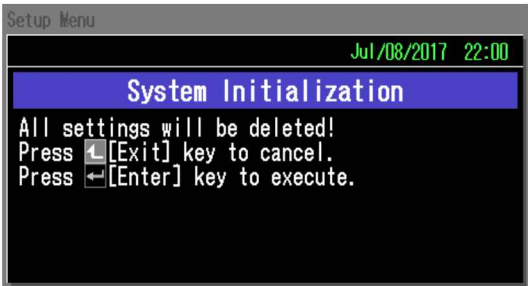
CAUTION Before starting the system initialization, be sure to save Common Settings and Preset Settings into the USB memory, by referring to "5.1.10 Setting Value Save/Recovery" or to the memory in the MCU board by referring to "5.1.11 Shipment setting".

TIP The setting data saved in the memory is not lost even if the system is initialized by the Level 3.

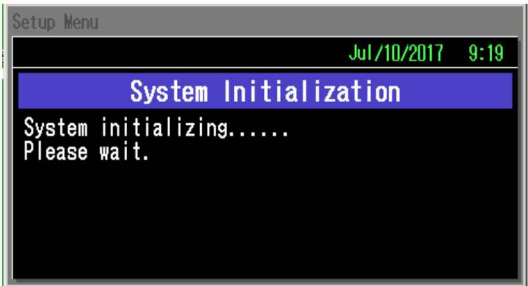
- 1. Point the cursor to [System Initialization], and press  .
• [System Initialization] menu will appear.



- 2. When performing the system initialization, press  .
Press  to return to the previous screen without initialization.



- [System Initialization] menu will appear.



- 3. When the initialization completes, returns to the [Setup menu] screen.

